

Censored multistate hazard (MSH) models

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1 Multi-state hazard models

The survival package deals with multistate hazard (MSH) models that are based on a non-parametric baseline hazard. The Aalen-Johansen estimator is a the natural multi-state extension of the Kaplan-Meier, and the MSH models themselves a natural extension of the Cox model. The `survfit` routine produces both the KM and AJ estimates, and the `coxph` function estimates both the simple proportional hazards (Cox) MSH model fits. We can think of these are MSH models based on observed transitions. A basic premise of the underlying likelihood is that, over the time window of observation for a subject, we directly observe all of the transitions made by that subject, i.e., what transitions and when.

The primary fitting function of this package, `cmsh`, deals with MSH models that are based on intermittently observed states. In the simplest case we will know that a particular transition must have occurred between patient visit k and $k + 1$ but won't know exactly when within that interval; the transition time is interval censored. In the most general case the current state at a visit can only be assessed indirectly, leading to a multistate hidden Markov model (MSHMM), the title for the package. (Footnote: we originally planned for two fitting functions, censored MSH = `cmsh` for the simple cases and `mshmm` for the more complex; but the arguments and computation for the two overlapped to such a great degree that one user level function now does both; we kept the simpler name.)

1.1 Colon cancer

A primary difference between models fit using `coxph` and `cmsh` is that the former is focused on modeling the *observed transitions*, and implicitly assumes we know when those transitions happened, while the latter on modeling the *observed state* at selected times. Consider Figure 1, for instance, in a cancer clinical trial where patients are followed up every 8 weeks. We could think of `cmsh` as providing a model for the time to *actual* progression and `coxph` a model for the time to *detection* of progression.

For a study with regular follow-up, such as a clinical trial, we have found that the results of a `coxph` and `cmsh` fit are often nearly identical, particularly with respect to covariates. If treatment decreases the rate of true progression by 30% it will do the same to the rate of detected progression.

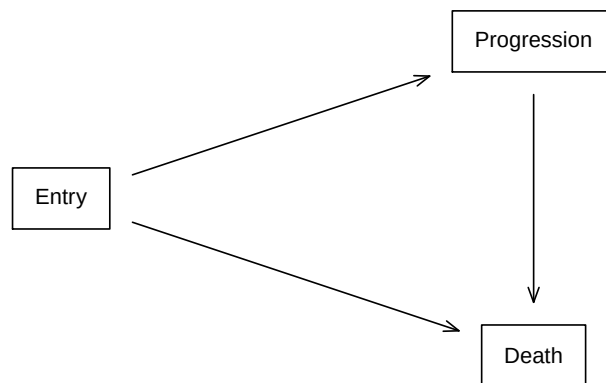


Figure 1: A simple model of cancer progression.

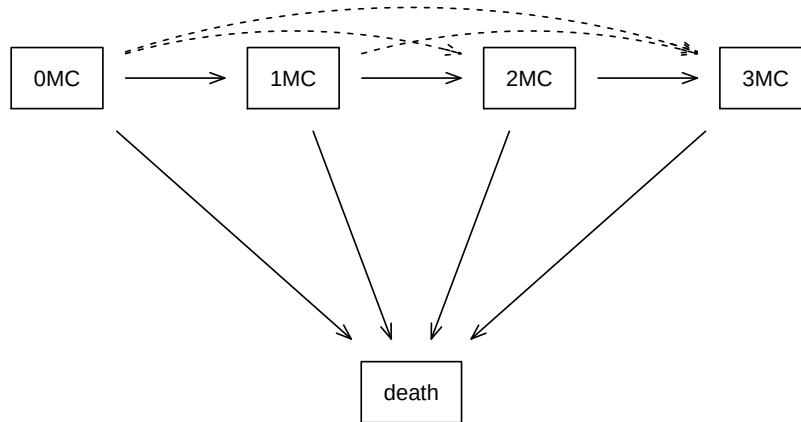


Figure 2: Transitions for the NAFLD data, 0–3 = number of the defined metabolid comorbidites.

1.2 Fatty liver disease

A more interesting case is a study of metabolic dysfunction-associated steatotic liver disease (MASLD), which we will use below in section 3.2. All adult subject subjects with a diagnosss of MAFLD from 1997 to 2012, along with 4 potential controls for each case, matched on age, sex, and year of office visit, were followed forward for the occurence of any of 3 metabolic comorbidities (MC): diabetes, hypertension, and hyperlipidemia. A state space is shown in Figure 2. (Since comorbidities in the study were defined as “history of diabetes”, “history of dyslipidemia”, and “history of hypertension” no arrows go backwards.) The follow-up visits are in the course of standard clinical care, and have wide differences in within and between subject spacing, with 20, 50, 80 and 95th percentiles of 3, 10, 19 and 50 months. If there are systematic differences in visit frequency, e.g., if women saw their physician more often on average than men, this can introduce a bias in the coxph analysis; the interval censored likelihood accounts for this variable spacing.

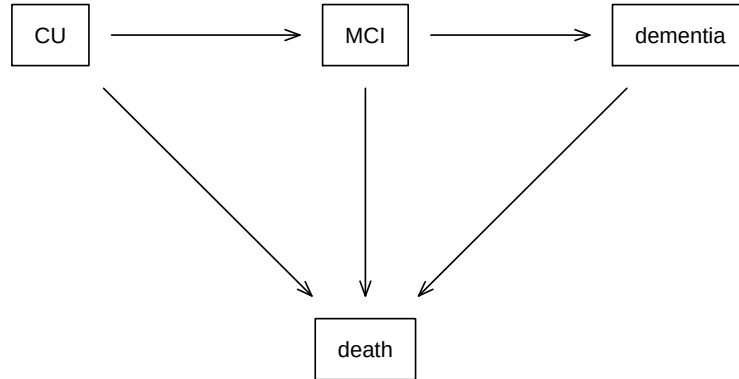


Figure 3: State space for progression of cognitive decline: states of cognitively unimpaired (CU), mild cognitive impairment (MCI), dementia and death.

1.3 Mayo Clinic Study of Aging

The Mayo Clinic Study of Aging is a population based study of cognitive change in older adults. Starting in 2004, an age and sex stratified sample of subjects aged 70–89 was recruited from Olmsted County, Minnesota. Subjects are scheduled to return at 15 month intervals, at each visit they are assessed with respect to cognitive status along with an array of other covariates. At regular intervals the cohort has been refreshed with new waves of recruitment, each wave widened the age range for recruitment. The **dementia** data set is a constructed data set based on a 2021 version of the MCSA data, with the same covariate, outcome, and visit distribution. The research needs of this study have been the motivation for most of the features in the package, and we will fit a variety of models using the dementia data. One simple example is shown in Figure 3.

1.4 Model and notation

One requirement for `cmsh` fits will be a transition matrix, e.g., the one below for the NAFLD study. It contains a positive value for any possible transition (usually 1), and 0 for transitions that are not. In the NAFLD data for instance, there are subjects whose have no comorbidities on their first visit and 2 on the next, which might be several years later. We know, biologically, that this subject had to pass through the 1MC state, but the program will not unless we tell it so.

	OMC	1MC	2MC	3MC	death
OMC	0	1	0.7	0.7	1
1MC	0	0	1.0	0.7	1
2MC	0	0	0.0	1.0	1
3MC	0	0	0.0	0.0	1
death	0	0	0.0	0.0	0

A second difference is that the `cmsh` routine uses a parametric baseline hazard for each state to state transition, most often a spline function of age or time. As a consequence the rates include an intercept term.

A third, a probably the most profound difference is that `cmsh` can accomodate partially observed outcomes. It turns out that there is a close connection between the simple interval censored multi-state models discussed so far, and multi-state hidden Markov models, i.e., one can fit them both using the same underlying function.

For some set of states we might observe only a set of one or more imperfect biomarkers. We might label stages of cerebral vascular damage as C0, C1, C2 and C3 for instance, where C0 is normal and C3 is severe. MRI based measures such as white matter hyperintensity (WMH) or fractional anisotropy can give partial insight into the current state, with direct measurement only available at autopsy.

An MSH model where all the state measures are indirect is known as a hidden Markov model: since the true state is never directly observed, the states are “hidden”. In the wikipedia entry on HMM for instance, refers to X as the states, Y the observed values and the density $f(Y|X)$ as the emission probability (or $P(Y \in A|X)$ for discrete Y). In our models we often have all three types of states

1. Transitions where both the state and the time of entry to the state are known, e.g., death.
2. Known states, where for a visit at time t the current state can be assessed accurately, but the time of entry to that state is not known.
3. Hidden (latent) states, which are measured only indirectly.

Since the majority of our models have been hybrids of the above, what one might call a semi-hidden Markov model, we have used `cmsh` (censored MSH) as the name of the primary fitting function rather than `hmm`. We will use ‘states’ to refer to true states, some of which may be latent, ‘markers’ for variables that are indirect measures of those states, ‘covariates’ for measured variables, e.g., sex, age, that affect the transition rates from state to state, the relationship between a marker and states, or initial state for each subject, and last a matrix of coefficients β . Then $\eta = X\beta$ is a matrix of linear predictors, each column mapping to some

aspect of the model, e.g., a particular transition rate. Here X is the standard n by p design matrix of statistics, one row per observation, containing intercept, covariate, spline term, etc columns.

We should state at the outset that these problems are also the target of the `msm` package, so why create a new one? There are currently four primary differences between the two packages.

1. From a user point of view, `mshmm` is designed to have similar structure and code to the survival package. The two packages share a primary author, who (of course) finds this style more natural.
2. For more complex data, like the dementia example found further below in this vignette, the underlying optimization functions in `msm` were very often not successful at finding the maximum, and very slow when they did succeed. Experimentation with more robust approaches, along with a need for these models within our dementia research program, have been the primary drivers for the `mshmm` package.
3. Follow-up summaries of the model, e.g., mean time in state, are more fully fleshed out in the `msm` package (though we plan to improve this).

As a user, 1 and 2 might recommend `mshmm`, while item 3, a longer history, and a deserved reputation for reliability would recommend `msm`. We emphatically do not claim that `mshmm` is superior. Use what works for you. Christopher Jackson, the author of `msm`, has provided invaluable feedback about technical aspects of the models, along with comments about fits and issues with particular data sets; both of which have been a key to the success of this endeavor.

2 Data

The `mshmm` functions use a different shape of input data than the multistate `survfit` and `coxph` functions in the survival package. The survival functions use data that has been cast into the counting process form:

- Each line has a subject id along with a time interval $(t1, t2]$ to which this observation applies.
- Covariates apply over the entire interval, from start to finish, likewise for stratification variables, if present.
- Outcomes occur at the end of the interval.

The data for `cmsh` should be in *timeline* form.

- Each line of the data has a subject id, along with a time value
- Other variables on the line are information that was measured or known *at that time point*

A timeline data set is actually more natural in the sense that it mimics the way in which data is gathered. It also may be easier to create, since one can use a wider variety of R tools. For ordinary survival, however, a Kaplan-Meier say, data in timeline form requires two rows of data per subject: the first observation at time 0 will have baseline covariates along with a status

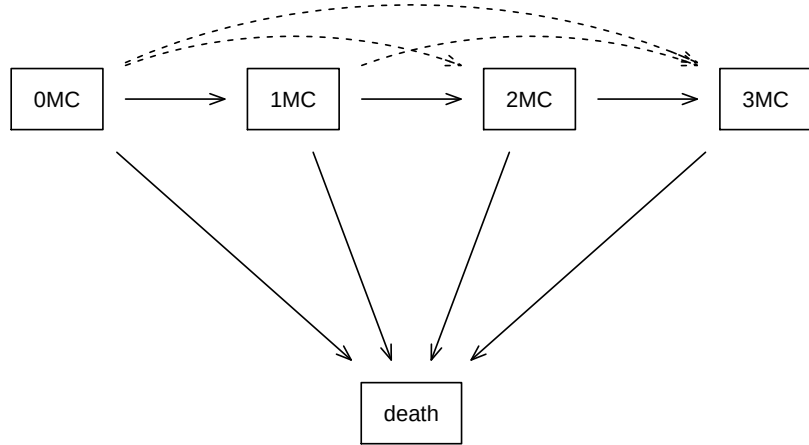


Figure 4: Observed transtions in the NAFLD data

of alive; a second observation at the time of death or last follow-up will have the status at that time point (alive/dead). This is different than the traditional way that survival data is coded, i.e., a single observation with a time/status variable pair. The multistate format for the survival package evolved out of simple survival and so retains that instance as a special case, i.e., one line per subject for the simple cases of alive/dead or competing risks.

A second observation is that a timeline dataset will have many missing values, e.g., blood pressure would not normally be known on the day of death, especially if death date is obtained from other sources. One consequence is that the usual first step of other modeling functions such as `lm` or `coxph`, i.e., to remove any data row that contains missing values, is not taken by the `cmsh` function. The handling of missings is more nuanced.

In interval censored multistate data, the subjects may be observed to be in some state s at time t , but the time of entry to that state is unknown. When there is interval censoring, the user will need to create a map of the valid transitions, as the underlying code is not able to do so with assurance. For example consider the data for the NAFLD study, found in the

survival package. Subjects were retrospectively categorized into one of 4 states at each office visit, with respect to the number of metabolic comorbidities (MC) that they had accumulated to that point; the three MC of interest are diabetes, hyperlipemia, and hypertension. Figure 4 shows the observed transitions. A given subject might have a multi-year gap between two office visits, leading to the accumulation of multiple MC in the interim; these observed transitions are drawn as dotted lines in Figure 4. The true transition matrix, which would be used as input to the `cmsh` function will omit the dashed lines; it has zero for transitions which do not actually occur (biologically).

In this data set the 1MC state is interval censored: for a subject who had none of the three conditions at their initial visit at age 56, say, and 2 MC observed at a visit 2 years later, we do not know exactly when that second condition arose in that interval. For almost all interval censored data sets there is at least one state which is not interval censored, i.e., death.

Contrast this to simpler multi-state hazard (MSH) model as fit by `coxph`, where the recorded times are taken to be the actual transition times. In the MSH model fit by `coxph` a subject is assumed to still occupy the state of their most recent transition until proven otherwise, e.g., observation of 1MC at 3 years and a second at 10 implies 1MC at 9.999...: there is an implicit assumption that all transitions are observed over the time interval. In an interval censored MSH, censoring means only “not dead at this time”. A subject who is 1MC at year 3 and censored at 10 could be in any of the 1MC, 2MC, or 3MC states at that later time.

Formally, the `cmsh` code treats a status of censored as “not in any state which is both exact and absorbing”. An exact state is one for which the time of entry to the state is known precisely, i.e., not interval censored. The list of exact states is provided by the `exact` argument; death is the most common example. Absorbing states can be inferred from the transitions matrix. It is possible to have an exact state which is not absorbing, an example is the surgical resection state in Crohn’s disease [11].

3 Interval censored data

3.1 Colon cancer trial

The `colon` data set in the survival package contains the results of a trial of three treatments for stage B/C colon cancer. This is a simple 3 state model of entry, progression, and death. The data set is organized with 2 rows per subject, the first containing the progression outcome and the second the death outcome. As a first step, however, we need to decide what to do with the 5 subjects for which progression and death were recorded on the same day. There is nothing wrong with this, per se: the author has for instance had studies where progression was discovered at autopsy, and one might reasonably expect an interval censored model to accommodate it. However, it has been our experience that “observation” of two states on the same day is more often a data creation error, and the `icmsh` routine currently flags it as such. To resolve the conflict we will shift those progressions one day earlier. (Or even .001 day if one wants.) Also beware the two for whom their progression date is also their last follow-up time, so as not to add a superfluous “censoring” row at the end.

```
temp1 <- subset(colon, etype==1) # progression
temp2 <- subset(colon, etype==2) # death
```

```

ties <- (temp1$time==temp2$time & temp1$status==1 & temp2$status==1)
temp1$time[ties] <- temp1$time[ties] -1

# Make the timeline data set, we will use only a few of the predictors
# The baseline data can use the row for either etype=1 or 2: the covariates
# are the identical
baseline <- subset(temp1, , c(id, rx, sex, age, node4, extent))
baseline$time <- 0
baseline$status <- -1
prog <- subset(temp1, status==1, c(id, time, status)) # progression
dfu <- subset(temp2, temp1$status==0 | time > temp1$time,
              c(id, time, status)) # death or last fu
dfu$status <- dfu$status *2 # recode as 0/2
temp3 <- merge(baseline, prog, all=TRUE)
cdata <- merge(temp3, dfu, all=TRUE)
cdata$status <- factor(cdata$status, c(0, -1, 1, 2),
                      c("censor", "stable", "progression", "death"))

subset(cdata, id<4)

```

	id	time	status	rx	sex	age	node4	extent
1	1	0	stable	Lev+5FU	1	43	1	3
2	1	968	progression	<NA>	NA	NA	NA	NA
3	1	1521	death	<NA>	NA	NA	NA	NA
4	2	0	stable	Lev+5FU	1	63	0	3
5	2	3087	censor	<NA>	NA	NA	NA	NA
6	3	0	stable	Obs	0	71	1	2
7	3	542	progression	<NA>	NA	NA	NA	NA
8	3	963	death	<NA>	NA	NA	NA	NA

Subject 1 has both progression and death, subject 2 is censored without progression at 3087 days.

One of the necessary conditions for a valid model, per underlying counting process theory, is the concept of a predictable process. In essence, an outcome observed at time t cannot depend on any predictors measured at or after time t . The `cmsh` function enforces this: any observed outcome on a given line of data is predicted using information from prior rows of data. One consequence is that prediction covariates on the final row for a subject have no impact. A second action within `cmsh` is to use the most recently observed value for any predictor; it is not necessary to fill in the values for treatment, age at entry, etc in every row of the data set. It is a bit of a semantic argument whether each row of a timeline data set should contain only values *measured* at that time or all values *known* at that time, i.e., whether fixed covariates like sex should be propagated across all rows. The `cmsh` function is happy with either, and more capable data set creation tools than the basic R merge used above may well fill these in.

As a first look compute the transition matrix. Then compute the probability in state and cumulative hazard curves. Many of the survival routines can accept timeline data via the `Surv2` operator.

```

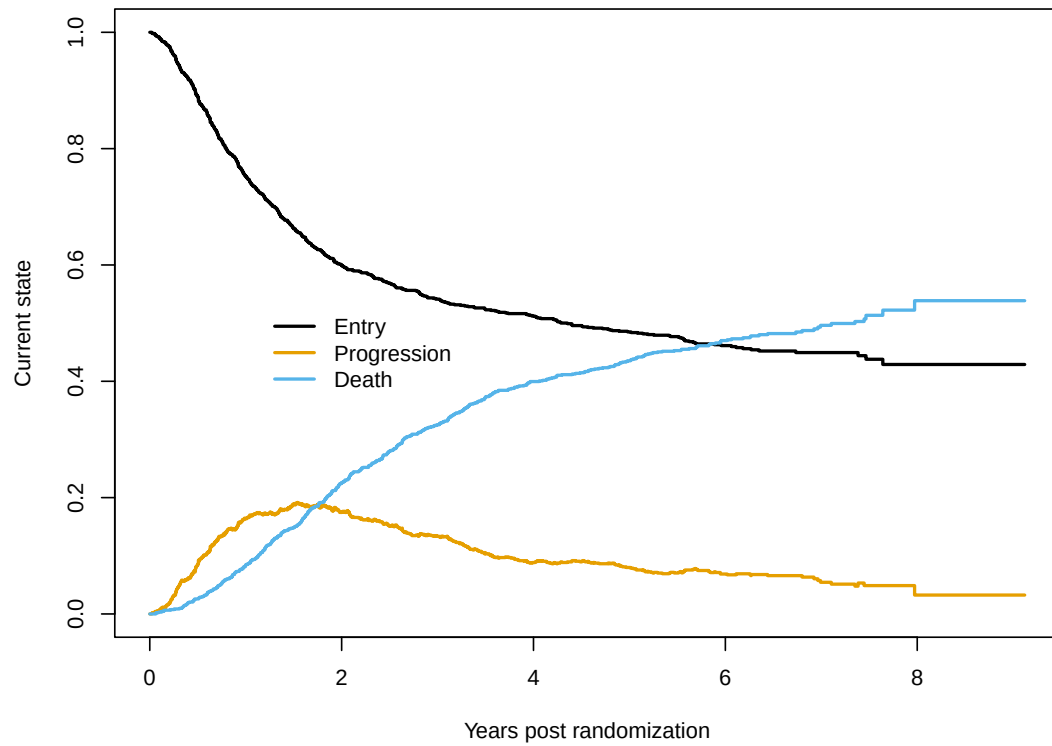
survcheck(Surv2(time, status) ~1, cdata, id=id)
Call:
survcheck(formula = Surv2(time, status) ~ 1, data = cdata, id = id)

Unique identifiers      Observations      Transitions
              929              1395              920

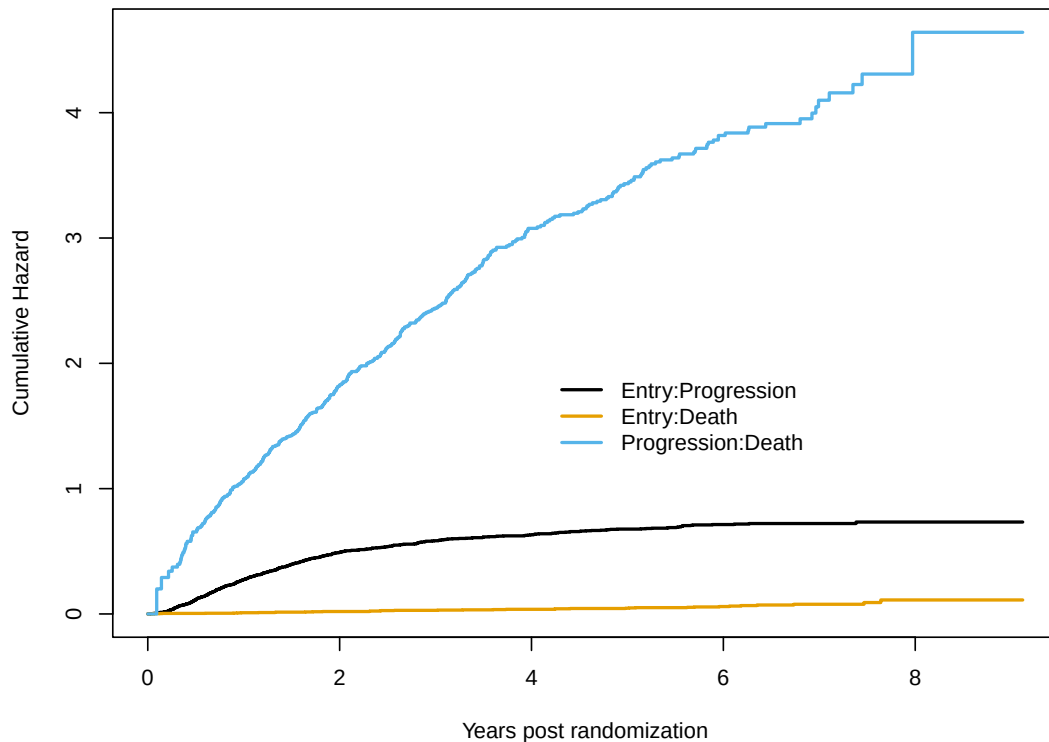
Transitions table:
      to
from   progression death (censored)
stable      468      38      423
progression      0     414      52
death         0      0      0

Number of subjects with 0, 1, ... transitions to each state:
      count
state    0   1   2
progression 461 468   0
death      477 452   0
(any)      423  92 414
aj1 <- survfit(Surv2(time, status) ~ 1, cdata, id=id)
plot(aj1, col=1:3, xscale=365, lwd=2,
      xlab="Years post randomization", ylab="Current state")
legend(400, .54, c("Entry", "Progression", "Death"), lty=1, lwd=2, col=1:3,
      bty='n')

```



```
plot(aj1, cumhaz=TRUE, col=1:3, xscale=365, lwd=2,
      xlab="Years post randomization", ylab= "Cumulative Hazard")
legend(1500, 2, c("Entry:Progression", "Entry:Death", "Progression:Death"),
      col=1:3, lwd=2, lty=1, bty='n')
```



In this study about 46% (423/929) of the subjects had no transitions, i.e., were still in the stable state at last observation, both death and progression was observed for 45% (414/929). Only a small number died without progression (38) or were in the progression state at last follow-up.

The cumulative hazards at 8 years are about .7, .1 and 4 for the transition rates of entry to progression, entry to death and progression to death, which provide reasonable starting estimates for the baseline hazards of the interval censored fit. Using years as the time scale will give more interpretable coefficients, so we add that variable.

Let us start with a simple Cox model fit to serve as comparison.

```
cfit0 <- coxph(Surv2(time, status) ~ rx, data=cdata, id=id)
cfit0
Call:
coxph(formula = Surv2(time, status) ~ rx, data = cdata, id = id)
```

	coef	exp(coef)	robust se	z	p
1:2					

```

rxLev      -0.01515    0.98497    0.10721 -0.141    0.888
rxLev+5FU  -0.51189    0.59936    0.11809 -4.335 1.46e-05

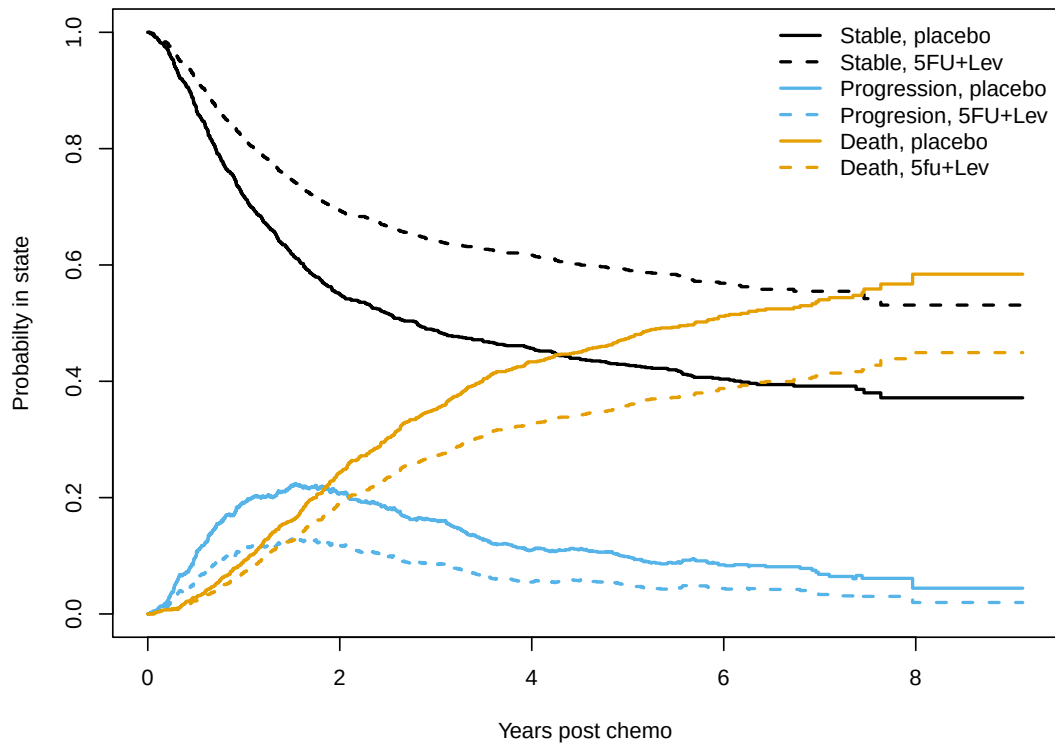
1:3          coef exp(coef) robust se      z      p
rxLev      -0.2839    0.7528    0.4204 -0.675 0.499
rxLev+5FU  -0.0878    0.9159    0.3822 -0.230 0.818

2:3          coef exp(coef) robust se      z      p
rxLev      0.04349    1.04445    0.10554 0.412 0.680
rxLev+5FU  0.27652    1.31853    0.12164 2.273 0.023

States:  1= stable, 2= progression, 3= death

Likelihood ratio test=29.89  on 6 df, p=4.124e-05
n= 1395, unique id= 929, number of events= 920
dummy <- data.frame(rx=c("Obs", "Lev+5FU"))
surv0 <- survfit(cfit0, newdata= dummy, p0=c(1,0,0))
dim(surv0)
  data states
    2      3
plot(surv0, lty=1:2, col=c(1,1,3,3,2,2), lwd=2, xscale=365.25,
     xlab="Years post chemo", ylab="Probabilty in state")
legend("topright", c("Stable, placebo", "Stable, 5FU+Lev",
                     "Progression, placebo", "Progresion, 5FU+Lev",
                     "Death, placebo", "Death, 5fu+Lev"),
     lty=1:2, lwd=2, col=c(1,1,3,3,2,2),, bty='n')

```

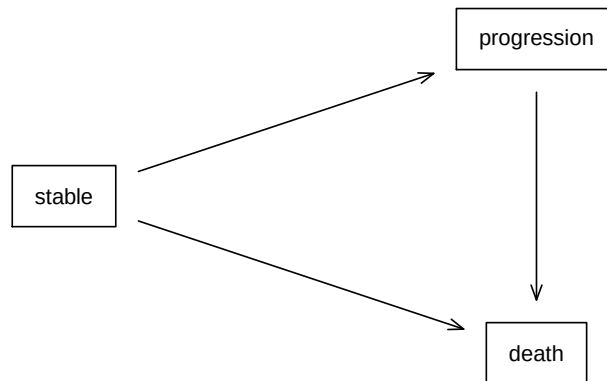


The hazard ratios show that combined treatment is more effective than either Levamisole alone or placebo with respect to progression, and the latter two are very similar. The predicted probability in state curves bear this out, and have a shape similar to the Aalen-Johansen. Since probability in state curves have to add to 1, we often leave out the “still in initial state” ones. Now create the fit and plot using interval censored.

```
# create the transition matrix
cstate <- c("stable", "progression", "death")
ctrans <- matrix(c(0,0,0,1,0,0,1,1,0), nrow=3,
                 dimnames= list(from= cstate, to= cstate))

ctrans

      to
from   stable progression death
stable      0          1      1
progression 0          0      1
death       0          0      0
statefig(1:2, ctrans) # visual check that it is correct
```



A transition matrix is necessary for the fit: possible transitions are marked by 1 and invalid transitions by 0. (The code actually allows any positive value, since the statefig routine uses non-integer values between 0 and 2 for drawing curved connections.)

```

# Crude transitions, as a starting estimate
crude <- c(.7, .1, 4)/(8*365)
cfit1 <- cmsh(Surv(time, status) ~ rx, id=id,
              init=rbind(log(crude),0,0),
              qmatrix= ctrans, exact="death", data= cdata,
              mc.cores=2)

cfit1
Call:
cmsh(formula = Surv(time, status) ~ rx, data = cdata, id = id,
      qmatrix = ctrans, init = rbind(log(crude), 0, 0), exact = "death",
      mc.cores = 2)

```

```

1:2      1:3      2:3

```

```

(Intercept) -2.786 -20.243 -8.00
rxLev       0.503  1.657 -0.04
rxLev+5FU   2.053  1.532 -0.40

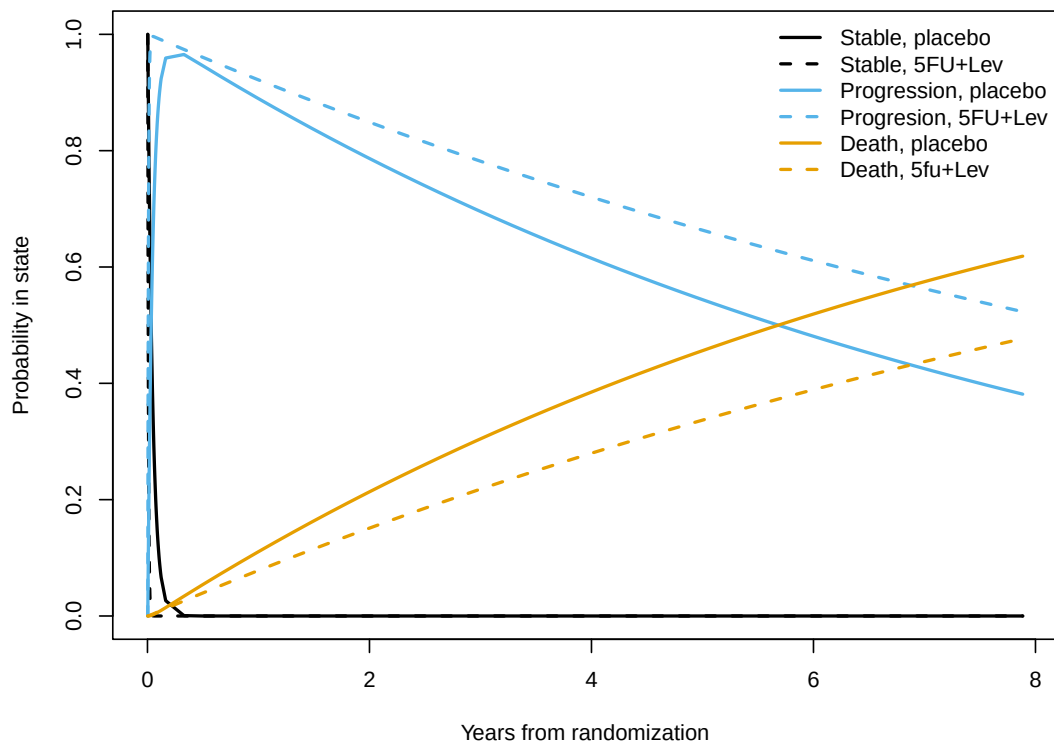
```

States: 1= stable, 2= progression, 3= death

```

Log-likelihood: ( -4930.603, -4123.282 ), 2*diff: 807.3 on 9 df
2324 observations 929 subjects
surv1 <- predict(cfit1, type="pstate", newdata=dummy, p0=c(1,0,0),
                 time= c(1:45, seq(60, 8*365, by=60)))
plot(surv1, lty=1:2, col=c(1,1,3,3,2,2), lwd=2, xscale=365.25,
     xlab="Years from randomization", ylab="Probability in state")
legend("topright", c("Stable, placebo", "Stable, 5FU+Lev",
                    "Progression, placebo", "Progression, 5FU+Lev",
                    "Death, placebo", "Death, 5fu+Lev"),
     lty=1:2, lwd=2, col=c(1,1,3,3,2,2),, bty='n')

```

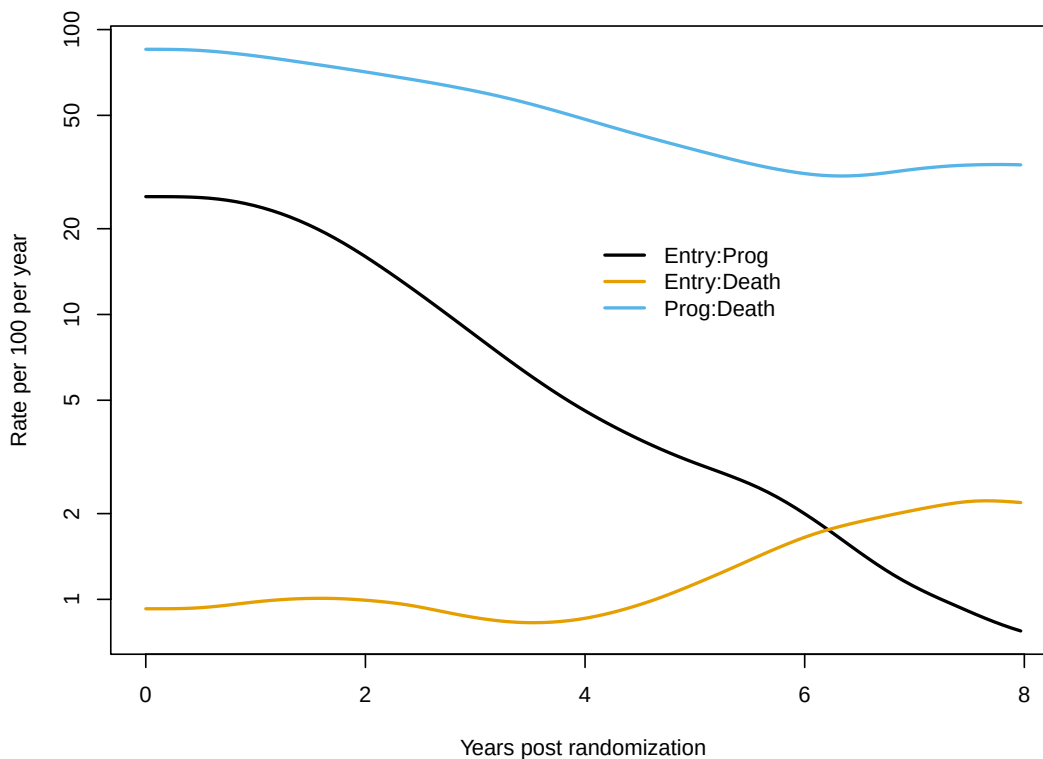


The cmsh solution is not at all what we expected. The estimated treatment coefficients are different than the Cox model and the predicted curves are *very* different: $P(\text{stable})$ drops almost immediately to zero. The issue is that we have no intermediate values in the data: there are no observations of “stable” after entry, and so the best MLE estimate for an interval censored model is to assume that subjects switch immediately to the progression state. The code iterates, increasing the stable:progression rate until the first progression, at 8 days, has a sufficiently large estimated $\text{Pr}(\text{progression})$ to declare convergence of the log-likelihood. This first progression on the 5FU+Lev arm, and that has the largest estimated 1:2 transition rate.

Per reference 1 in the help file for the data, patients were assessed for progression every 8 weeks for the first year, every 4 months for the second year, and every 6 months for years 3 to 5. To properly analyse this data using interval censored methods, data rows for each of these intermediate visits should be included in the study data, but this data is not available to us. An interesting exercise is to add psuedo observations that match the visit schedule, and observe how the estimate behaves. Since subjects cannot revert from progression, it suffices to add a single pseudo visit prior each progression, e.g., id 1 had observed progression at 968 days = 31.7 months so assume a non-progression visit 6 months before at 786 days. (Or perhaps seven months, since people are more often late than early). For progressions after 5 years we will assume yearly visits.

However, that is not enough. The multistate Nelson-Aalen hazard estimate shown earlier does not look like constant hazard. A somewhat ad hoc way to show the hazard instead of the cumulative hazard is to fit it using the `smooth.spline` and then plot the first derivative of the fit.

```
xtime <- seq(0, 8*365, by=30)
scurve <- sapply(1:3, function(i) {
  smooth <- smooth.spline(aj1$time, aj1$cumhaz[,i], df=5)
  predict(smooth, xtime, deriv=1)$y
})
matplot(xtime/365.25, scurve*365.25, type='l', lwd=2, col=1:3, log='y', lty=1,
        xlab="Years post randomization", ylab="Rate per 100 per year")
legend(4,20, c("Entry:Prog", "Entry:Death", "Prog:Death"), lty=1, lwd=2,
       col=1:3, bty='n')
```



Since rates in the `cmsh` and `coxph` function are $\lambda(t) \exp(X\beta)$ we see that using a linear term $\beta_0 + \beta_1 t$ for the baseline could suffice, a spline would be more flexible. This leads to an overall hazard of

$$\lambda(x, t) = \exp(\beta_0 + \beta_1 t + \beta_2 I(trt = 2) + \beta_3 I(trt = 3))$$

In an `mshmm` fit the underlying hazard, as a function of time, is an explicit part of the model. This is unlike `coxph` where the baseline hazard function is implicit, and estimated non-parametrically after solving for $\hat{\beta}$. For this data, treating the rate as a constant over time is not a satisfactory fit. This was implicitly assumed in `cfit1` above, but that fit had bigger problems.

Years post follow-up is a time-dependent variable which we will need to explicitly manufacture and add to the data set. Yet to do...

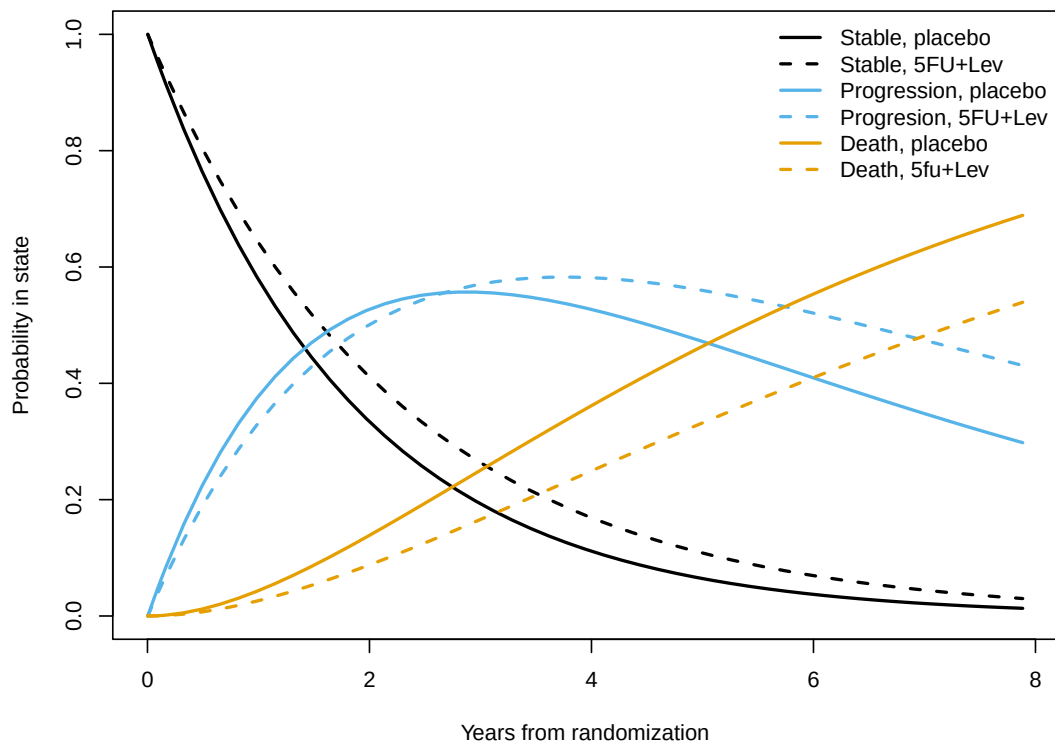
```
# assume progression in the first 10 weeks had no prior 'clear' visit
tdata <- subset(cdata, status=="progression" & time > 80)
pgroup <- cut(tdata$time, c(0, 80, 365+56, 730+92, 5*365+ 182, max(tdata$time)))
pgroup <- as.numeric(pgroup)
add <- subset(tdata, , c(id, time, status))
```

```

add$time <- add$time- c(0, 57, 125, 190, 370)[pgroup]
add$status <- "stable"
cdata2 <- merge(cdata, add, all=TRUE)

cfit2 <- cmsh(Surv(time, status) ~ rx, id=id,
              init=rbind(log(crude),0,0),
              qmatrix= ctrans, exact="death", data= cdata2,
              mc.cores=2)
surv2 <- predict(cfit2, type="pstate", newdata=dummy, p0=c(1,0,0),
                 time=c(1:45, seq(60, 8*365, by=60)))
plot(surv2, lty=1:2, col=c(1,1,3,3,2,2), lwd=2, xscale=365.25,
     xlab="Years from randomization", ylab="Probability in state")
legend("topright", c("Stable, placebo", "Stable, 5FU+Lev",
                     "Progression, placebo", "Progression, 5FU+Lev",
                     "Death, placebo", "Death, 5fu+Lev"),
      lty=1:2, lwd=2, col=c(1,1,3,3,2,2),, bty='n')

```



The remainder is not filled in yet (volunteers?)

- Based in the cumulative hazard plot the years on study effect may need to be nonlinear.
- Compare results and curves to a multistate Cox for the same data
- My expectation is almost no change in hazard ratio coefficients, and the time to treatment failure estimate shifted 1-2 months earlier.
- Message: when visits are at regular intervals, there is not a great deal of clinical difference between the two endpoints of “time to observed progression” and “time to biological progression”; any covariate that affects one will have a similar effect on the other. The analysis will reflect that.

3.2 Fatty liver disease

Metabolic dysfunction-associated fatty liver disease or MASLD (previously known as NAFLD) is characterized by fat accumulation in the liver, and is linked to metabolic dysfunction, such as obesity, type 2 diabetes, or other signs of metabolic dysregulation. Allen [2] collected information on all adult subject subjects with a diagnosis of MASLD from 1997 to 2012, along with 4 potential controls for each case, matched on age, sex, and year of visit. These were followed forward through the medical history for the occurrence of any of 3 metabolic comorbidities (MC): diabetes, hypertension, and hyperlipidemia. A reprise of that analysis, based on the multistate model found in Figure 5, is found in the survival package.

If measured continuously, subjects would progress via the solid arrows from 0 to 1 to 2 to 3 comorbidities; we assume that no one progresses on two of the markers at the *exact* same instant. However, the interval between clinical visits varies considerably from subject to subject and visit to visit, with 20, 50 and 80th percentiles of 3, 9, and 18 months (max 245). If there are systematic differences in visit frequency, e.g., if women saw their physician more often on average than men, this can introduce a bias in the coxph analysis and cmsh may be more appropriate for that reason alone.

In the observed data, there will be subjects who are in the 0MC state at one visit, and in the 2MC state at their next visit; as that following visit might be several years later. Such visit pairs are shown as dashed lines in Figure 2. A fit based on observed transitions, using coxph, will model all observed transitions, including these jumps. One based on the interval censored model will model only the solid arrows, i.e., the transitions that actually occur. (Footnote: since comorbidities in the study were defined as “history of diabetes”, “history of dyslipidemia”, and “history of hypertension” no arrows go backwards.)

Like the colon cancer example above we need to include intermediate visits. The survival package’s NAFLD data consists of 3 data sets: baseline information with one row per subject (`nafld1`), outcome information (`nafld3`), and a third with selected office and/or laboratory values for each visit. In order to construct the interval censored data we will assume that each office visit represents a potential observation time for progression of the comorbidities. The metabolic comorbidity status at a subsequent death/last known alive date is not known, however.

Now create the timeline data set. For illustration we have done this using only the functions in base R, but several other, more sophisticated tools can be used. A couple of notes:

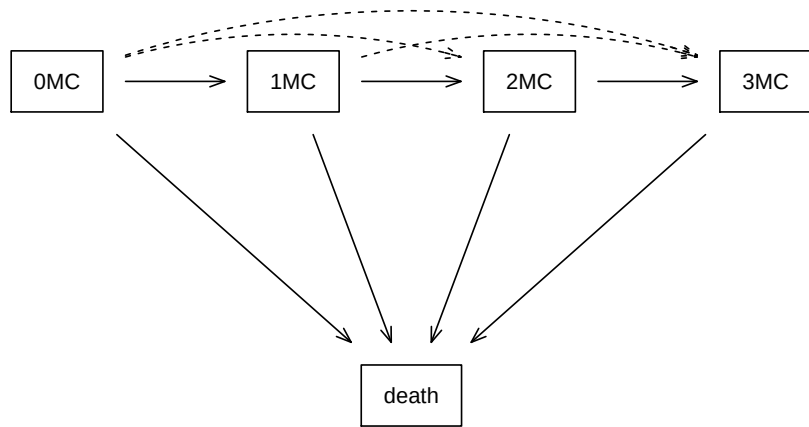


Figure 5: Observed transitions for the MASLD data

- The code `index <- match(id, id)` is an old programming trick in R (and in S-Plus and S before that). It creates an index that points each row of data to the first row with that id value.
- In general, in a timeline data set the first row for each subject will have few if any missings.

```
# baseline visit
base <- subset(nafld1, select=c(id, age, male, case.id))
base$days <- 0
base$nafld <- with(base, 1*(!is.na(case.id) & id==case.id)) #nafld at baseline
base$case.id <- NULL # no longer needed

# cumulative MC count
# times <=0 in nafld3 are conditions found at or before enrollment date
# create the "number of comorbidities so far" variable
temp2 <- subset(nafld3, event %in% c("diabetes", "dyslipidemia", "htn"),
               c(id, days)) # only events of interest
temp2$days <- pmax(0, temp2$days) # set prior events to day 0
#cumulative count
findex <- match(temp2$id, temp2$id)
temp2$mc <- 1+ 1:nrow(temp2) - (1:nrow(temp2))[findex]
# keep the last of any rows that are tied on id and days (2 MC on same visit)
temp2 <- temp2[!duplicated(temp2[,1:2], fromLast=TRUE),]

lfu <- subset(nafld1, select= c(id, futime, status)) # death or last fu
names(lfu)[2] <- "days"

ndata1 <- merge(base, temp2, by=c("id", "days"), all=TRUE) # other variables
ndata1 <- merge(ndata1, lfu, by=c("id", "days"), all=TRUE)# add futime

# everyone needs a starting MC state
ndata1$mc[!duplicated(ndata1$id) & is.na(ndata1$mc)] <- 0
# ndata1 is a timeline data
# manufacture the state variable, as a factor suitable for Surv
temp <- with(ndata1, ifelse(!is.na(mc), mc+1,
                          ifelse(!is.na(status) & status==1, 5, 0)))
ndata1$state <-factor(temp, 0:5,
                    c("censor", "0MC", "1MC", "2MC", "3MC", "death"))

# print out the transition matrix
survcheck(Surv2(days, state) ~1, ndata1, id=id)$ transitions
to
from      1MC  2MC  3MC death (censored)
0MC      1829   70    4   263         5705
1MC         0 1843   28   243         4567
```

```

2MC      0    0 1049   416      3687
3MC      0    0    0   441      2220
death    0    0    0    0        0
# Add in the office visits. There will occasionally be labs on sequential
# days for the same visit, but it isn't worth removing these "duplicate" visits.
# But do remove two tests on the same day.
office <- subset(nafld2, days>0, c(id,days))
office <- office[!duplicated(office),] # tests on the same day
ndata2 <- merge(ndata1, office, by=c("id", "days"), all=TRUE)
ndata2$state <- lvcf(ndata2$id, ndata2$state)

```

```

# The new transition table
survcheck(Surv2(days, state, repeated=TRUE) ~1, ndata2, id=id)$transitions
to
from      OMC    1MC    2MC    3MC death (censored)
OMC      8584   1829    70     4    263      5705
1MC       0  21283   1843    28    243      4567
2MC       0     0  25793   1049   416      3687
3MC       0     0     0  22631   441      2220
death     0     0     0     0     0        0

```

We see that data set `ndata1` has the exact same off-diagonal counts as the expanded data set `ndata2`, this is as it should be. The second data frame adds `0MC:0MC`, `1MC:2MC`, `2MC:2MC` and `3MC:3MC` transitions, all of which serve to narrow the time range over which a given transition between states will have occurred. A printout for id 1 shows `0MC` visits at 0, 183, 2030, 3578 and 5420 days. Because there are no backwards transitions we could delete the middle rows, as `0MC` at day 0 and `0MC` at day 5420 implies `0MC` throughout the interim. The likelihood will be the same without the middle rows, and computation a smidge faster. Deletion is not valid, however, if `0MC:1MC` and `1MC:0MC` transitions were possible. Think hard before thinning a data set.

```

# Actually, there are a lot of unneeded rows. Since there are not any
# TD covariates it is easy to count: the first row of a repeated pair adds
# no further information (except the first row)
isdup <- with(ndata2, diff(id)==0 & diff(as.numeric(state))==0)
isdup <- c(isdup,FALSE) & duplicated(ndata2$id)
table(isdup)
isdup
FALSE TRUE
52026 66179
ndata2 <- ndata2[!isdup,]

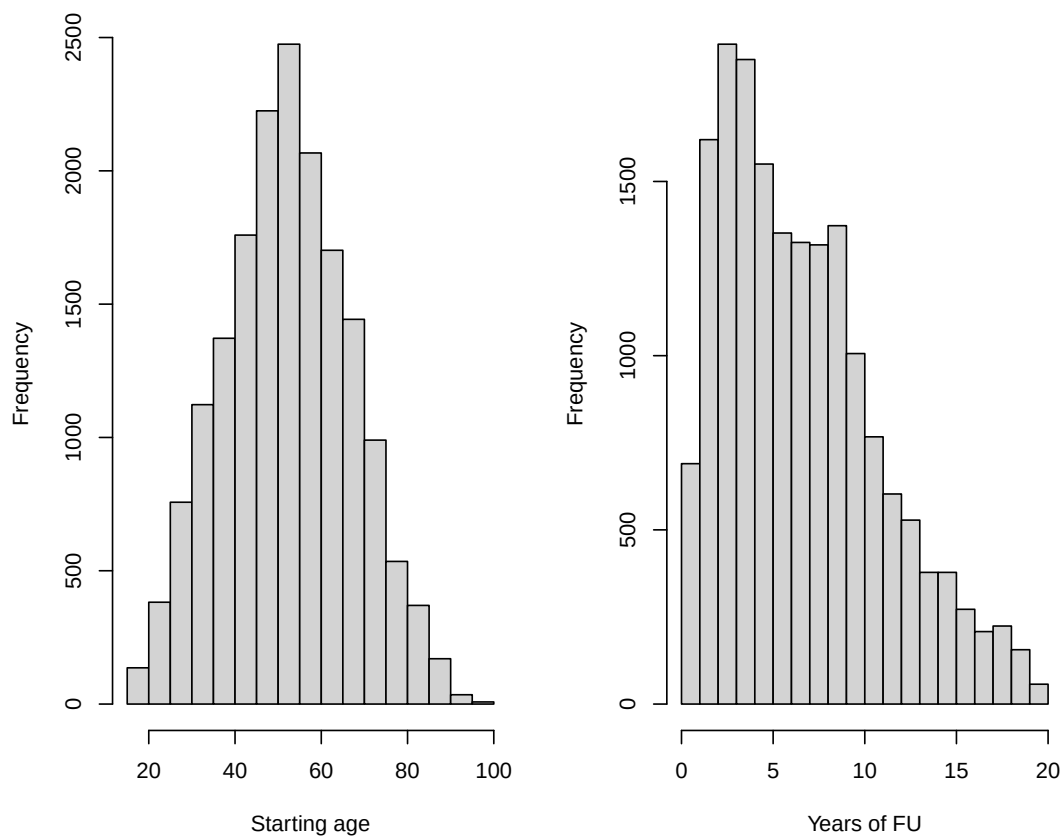
```

As is often the case, the above seemingly simple section of code to create the analysis data set required more time and effort than the analysis that appears below. It is imperative to create the table of transitions, look at it, and think about it. The row for death for instance should always be 0. (Make sure to include missings in any tables, we defined a `table2` function at

the start of this vignette to avoid having to type the `useNA= 'ifany'` option again and again.) Print out final data for a half dozen subjects and READ it.

One more step is needed, which is to add current age as a time-dependent covariate. The underlying rate of MC rises very rapidly with age, and this data set covers a wide range. The figure below shows the distributions of initial age and years of follow-up. Using only baseline age as a covariate would imply that for two subjects, both currently age 85, one who entered the study at age 70 and the other at age 80, that the current death rate is lower for the earlier recruit. For the coxph model we are able to use age as the time scale, which causes this careful matching to happen silently behind the scenes, but the cmsh routine has a parametric baseline and needs an explicit current age variable.

Why did we choose to make 1 year age intervals? Several centuries of experience with population rate tables, e.g., the US death rates found in `survival::survexp.us` or more generally those available from the Human Mortality Database (www.mortality.org) have shown that, other than for infants, yearly event rates provide a sufficiently fine division of age. For this data a coarser grid would almost certainly be acceptable, e.g., every 5 years of age, but we have chosen the classic strategy.



```

# create a dummy data set with integer ages
# (pretend they had shown up each birthday, but we only recorded "alive")
account <- floor(nafld1$futime/365.25) # number of extra ages to add
itemp <- unlist(lapply(account, function(i) 0:i))
dummy <- data.frame(id= rep(nafld1$id, 1+ account), days=round(itemp*365.25),
                    iage = itemp)

# merge it in
ndata3 <- merge(ndata2, dummy, by=c("id", "days"), all=TRUE)
ndata3$state[is.na(ndata3$state)] <- "censor"
ndata3$age <- lvcf(ndata3$id, ndata3$age)
ndata3$iage<- lvcf(ndata3$id, ndata3$age + ndata3$iage)
subset(ndata3, id==4)

```

	id	days	age	male	nafld	mc	status	state	iage
30	4	0	56	1	0	2	NA	2MC	56
31	4	365	56	NA	NA	NA	NA	censor	57
32	4	730	56	NA	NA	NA	NA	censor	58
33	4	1096	56	NA	NA	NA	NA	censor	59
34	4	1461	56	NA	NA	NA	NA	censor	60
35	4	1826	56	NA	NA	NA	NA	censor	61
36	4	2192	56	NA	NA	NA	NA	censor	62
37	4	2557	56	NA	NA	NA	NA	censor	63
38	4	2912	56	NA	NA	NA	NA	2MC	63
39	4	2922	56	NA	NA	NA	NA	censor	64
40	4	3143	56	NA	NA	NA	0	censor	64

The resulting data set has rows for both the visits, which have a state of 0MC–MC3, rows for the deaths, and rows for each age change, which have the generic state “censor” since the true state was not observed at that point. Current age is a time-dependent covariate. We will discuss issues with general time-dependent covariates later, but age and time since enrollment are much simpler than others, since their value at any time is known a priori, without requiring measurement at a patient visit. The reason for integer age will appear at the first theory discussion.

As a final check of the data set use it to create Aalen-Johansen curves. The Aalen-Johansen fit does not account for interval censoring, so can use `Surv2` and `ndata1`, the data before adding the office visits or time-dependent age. Below is the code which generates figure 6. The resulting computation will have hazards for both the solid and dashed line transitions of Figure 5, but we have chosen to display only the solid ones. The right hand panel has a line for each state. The hazard plot shows higher transition rates for the MC transitions than for death, and rough linearity. The $p(t)$ plot shows the gradual cascade through 0–3 MC. As stated just above, a primary purpose at this point was as a data check.

```

ndata3$years <- ndata3$days/365.25
ndata1$years <- ndata1$days/365.25

# create the current age variable
index <- match(ndata1$id, ndata1$id)

```

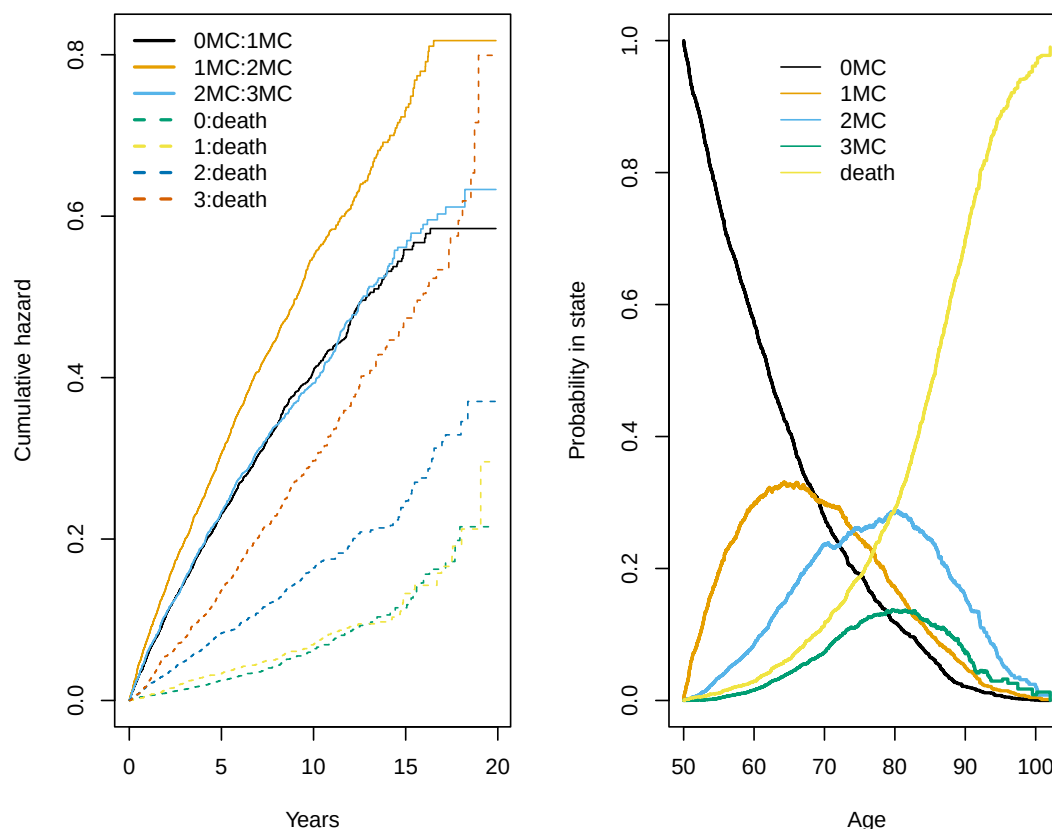


Figure 6: Left panel: estimates of the cumulative hazard for selected transitions. Right panel: estimated $p(t)$ for subject starting in the 0MC state at age 50.

```
ndata1$age <- ndata1$age[index] + ndata1$years

ajfit1 <- survfit(Surv2(years, state) ~ 1, ndata1, id=id)
#age scale
ajfit2 <- survfit(Surv2(age, state) ~1, ndata1, id=id, start.time=50,
                  p0=c(1,0,0,0,0))
```

For the actual model fit we need initial values for the parameters. I have found to my dismay that I don't do a very good job of picking initial coefficients when there are covariates. The estimates actually don't need to be that good, just "in the ballpark", but it is easy to create starting values that give crazy per-subject starting rates. A good idea appears to start with a model that has only the intercept, or intercept + age, either of which might be thought of as the

“null model”. For the former we can use the slope of a cumulative hazard plot such as figure 6, for the latter a simple Poisson glm model for death has been effective.

```
gdata <- data.frame(ndata3,
                    delta = diff(c(0, ndata3$years)))
gdata <- subset(gdata, duplicated(id))
glmfit <- glm(state=='death' ~ iage + offset(log(delta)), poisson, data=gdata)
icoef <- outer(coef(glmfit), rep(1,7), '*')
# optional refinement based on the figure, MC:MC rates are 3x death
# icoef[1,1:3] <- icoef[1, 1:3] + log(3)

states <- c(paste0(0:3, "MC"), "death")
nmat2 <- matrix(0,5,5, dimnames= list(states, states))
nmat2[1,2] <- nmat2[2,3] <- nmat2[3,4] <- 1
nmat2[1:4,5] <- 1
nmat2
      OMC 1MC 2MC 3MC death
OMC      0   1   0   0     1
1MC      0   0   1   0     1
2MC      0   0   0   1     1
3MC      0   0   0   0     1
death    0   0   0   0     0
nfit0 <- cmsh(Surv(years, state) ~ iage, ndata3, id= id,
              qmatrix= nmat2, init= icoef)
nfit0
Call:
cmsh(formula = Surv(years, state) ~ iage, data = ndata3, id = id,
      qmatrix = nmat2, init = icoef)

              1:2      2:3      3:4      1:5
(Intercept) -4.09317 -4.15242 -3.11593 -11.77989
iage         0.03224  0.02692  0.00391  0.10913
              2:5      3:5      4:5
(Intercept) -11.09970 -10.82008 -8.71
iage         0.09539  0.09533  0.08

States:  1= OMC, 2= 1MC, 3= 2MC, 4= 3MC, 5= death

Log-likelihood: ( -27222.76, -18609.96 ), 2*diff: 8612.8 on 14 df
159060 observations 17549 subjects
nfit1 <- cmsh(list(Surv(years, state) ~ iage + male + nafld,
                  0:5 ~ male/common),
              ndata3, id=id, qmatrix=nmat2, init=nfit0)
nfit1
Call:
cmsh(formula = list(Surv(years, state) ~ iage + male + nafld,
```

```
0:5 ~ male/common), data = ndata3, id = id, qmatrix = nmat2,
init = nfit0)
```

	1:2	2:3	3:4	1:5
(Intercept)	-4.40745	-4.58904	-3.66443	-12.08682
iage	0.03452	0.03068	0.00889	0.11097
male	0.27603	0.24779	0.19485	0.31937
nafld	0.52444	0.40633	0.40881	0.64824

	2:5	3:5	4:5
(Intercept)	-11.40565	-11.37030	-8.88
iage	0.09696	0.09907	0.08
male	0.31937	0.31937	0.32
nafld	0.35915	0.51492	0.00

States: 1= OMC, 2= 1MC, 3= 2MC, 4= 3MC, 5= death

Log-likelihood: (-18609.96, -18463.28), 2*diff: 146.7 on 25 df
159060 observations 17549 subjects

Now compare this to the MSH fit treating transition times as exact.

```
nfit2 <- coxph(list(Surv2(age, state) ~ male + nafld,
                   0:5 ~ male/common,
                   1:2 + 1:3 + 1:4 ~ male + nafld/common,
                   2:3 + 2:4 ~ male + nafld/common),
              ndata1, id=id)
nfit2
Call:
coxph(formula = list(Surv2(age, state) ~ male + nafld, 0:5 ~
  male/common, 1:2 + 1:3 + 1:4 ~ male + nafld/common, 2:3 +
  2:4 ~ male + nafld/common), data = ndata1, id = id)
```

1:2, 1:3, 1:4	coef	exp(coef)	robust se	z	p
male	0.18457	1.20270	0.04819	3.83	0.000128
nafld	0.93665	2.55142	0.06903	13.57	< 2e-16

2:3, 2:4	coef	exp(coef)	robust se	z	p
male	0.24910	1.28286	0.04987	4.994	5.9e-07
nafld	0.51520	1.67397	0.05752	8.957	< 2e-16

3:4	coef	exp(coef)	robust se	z	p
male	0.14672	1.15803	0.06501	2.257	0.024

```

nafld 0.46007    1.58418    0.06912 6.657 2.8e-11

1:5      coef exp(coef) robust se      z      p
male 0.3292    1.3898    0.0547 6.017 1.77e-09
nafld 0.7236    2.0617    0.2364 3.061 0.00221

2:5      coef exp(coef) robust se      z      p
male 0.3292    1.3898    0.0547 6.017 1.77e-09
nafld 0.5528    1.7382    0.1524 3.627 0.000287

3:5      coef exp(coef) robust se      z      p
male 0.3292    1.3898    0.0547 6.017 1.77e-09
nafld 0.5222    1.6858    0.1094 4.772 1.82e-06

4:5      coef exp(coef) robust se      z      p
male 0.32918    1.38983    0.05470 6.017 1.77e-09
nafld 0.04679    1.04790    0.10020 0.467    0.641

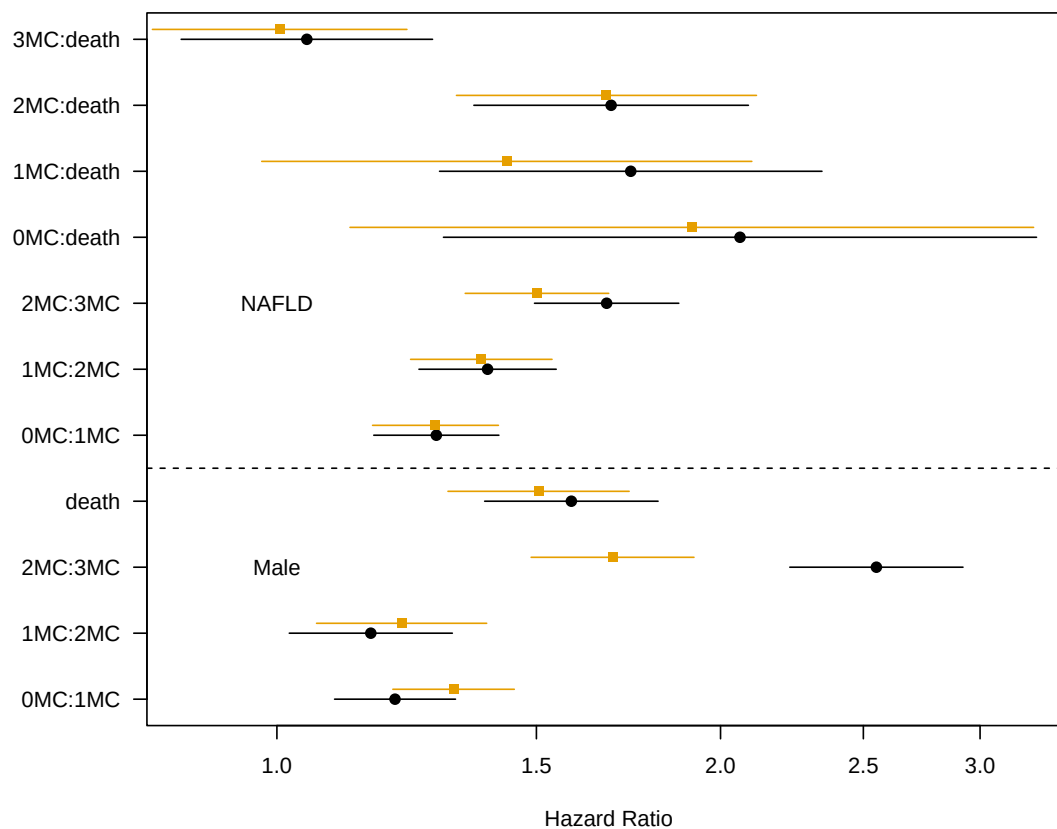
States:  1= OMC, 2= 1MC, 3= 2MC, 4= 3MC, 5= death

Likelihood ratio test=431.6  on 11 df, p=< 2.2e-16
n= 22365, unique id= 17549, number of events= 6186
# Hazard ratio plot
xx <- summary(nfit2)$conf.int[, -2]
yy <- c(1,3,5,7, 2,4,6,8:11)
opar <- par(mar=c(5,6,1,1))
matplot(xx, yy, log='x', type='n', xlab="Hazard Ratio", ylab="", yaxt='n')
points(xx[,1], yy, pch=19) #, col=rep(2:1, c(4,7))[yy])
segments(xx[,2], yy, xx[,3], yy)
abline(h=4.5, lty=2)
text(c(1, 1), c(3, 7), c("Male", "NAFLD"))
nstate <- nfit1$states
temp <- c(paste(nstate[1:3], nstate[2:4], sep=":"))
ylab <- c(temp, "death", temp, paste(nstate[1:4], "death", sep=":"))
axis(2, 1:11, ylab, las=2)

# I don't want the intercept or age coeffs
index <- unique(c(nfit1$cmap[3:4,]))
tempse <- sqrt(diag(solve(nfit1$fit$S2))) # theory is still weak
xx2 <- exp(coef(nfit1)[index] + outer(tempse[index], c(0, -1.96, 1.96), '*'))
delta <- .15

```

```
points(xx2[,1], yy+ delta, col=2, pch=15)
segments(xx2[,2], yy+delta, xx2[,3], yy+delta, col=2)
```



```
par(opar)
```

Yet to do:

- Note that nfit0 can be slow: initial estimates are hard
- Compare predicted $p(t)$ and $E(\text{time in state})$ estimates
- Come to some conclusions

4 Theory, pass 1

A big difference between `coxph` and `cmsh` is how the likelihood is constructed. The semiparametric MSH (`coxph`) is a model for observed *transitions*, while the interval censored (`cmsh`)

approach is a model for observed *states*. The semiparametric model assumes that we see all of the transitions, up to the point they are censored. It uses those transition and their timings, via risk sets, to construct estimated rates. These rates can then be used to compute probability in state curves $p(t)$.

Let $R(t)$ be the multistate rate matrix at time t . Off diagonal element $R_{ij}(t; x) = \hat{\lambda}_{ij}(t; x)$ contains the estimate transition rate between state i and state j , at time t , for a subject with covariate vector x . To normalize R set its diagonal element such that row sums are 0. (This is just what has to be done to compute the matrix exponential, no serious theory is involved wrt the normalization step.) If we assume a set of time points $t_0, t_1, t_2, \dots$ such that the transition rates are constant over each interval $t_i \leq t < t_{i+1}$ the the probability in estimate satisfies

$$p(t_i; x) = p(t_0) e^{(t_1 - t_0)R(t_0; x)} e^{(t_2 - t_1)R(t_1; x)} \dots e^{(t_i - t_{i-1})R(t_{i-1}; x)} \quad (1)$$

where $p(t)$ is a vector with one element per state, $p_k(t)$ the probability of being in state k , and $p(t_0)$ is the probability vector for the starting state, normally all zeros with a single “1” corresponding to the starting state. The exponential of a matrix, e.g. $\exp(R)$, is dealt with at length in the mathematics vignette. Suffice it to say that computation of this quantity, and of its derivative, is more complex than the exponential of a single value [7, 8, 1].

In the semiparametric MSH first the parameter estimates $\hat{\beta}$ are computed based on a partial likelihood, predictions $p(t; z)$ for any chosen covariate vector z are computed at the very end. In the interval censored MSH $p(t; x)$ is computed during maximization, that is, for each trial solution of $\hat{\beta}$ and each subject i

1. Compute the predicted curve $p(t; x_i)$ for the subject. For the moment assume the covariate $x_i(t)$ is not time-dependent. That will be a later topic.
2. Compare the prediction to the observed state for subject i , at each observation time t_{ij} for that subject.
3. Use this to compute an updated $\hat{\beta}$.

The computation of a likelihood in `cmsh` is a sum over subjects, and each subject can be computed independently. (We take advantage of this and use parallel computation if possible, either set `mc.cores` using `options` or include it in the `cmsh` call.) The algorithm for the likelihood of each subject alternates two steps. First step forward to a new time point using the matrix exponential as in equation (1). Whenever a subject is observed, restrict their $p(t)$ by setting to 0 all of the “states where they ain’t”, i.e., all elements of p except the observed state. The mathematics vignette has more formal details.

5 Partially observed states

The simplest extension of the simple interval censored model, and the simplest hidden Markov model, involves a partially observed state. An example appears in the Mayo Clinic Study of Aging, represented in the `dementia` data set. Jack et al [3] looked at a simple model of time to detection of MCI and/or dementia, as shown in figure 7. However, for some visits the formal cognitive testing necessary to distinguish mild cognitive impairment (MCI) is not available, at those visits subjects are classified as non-demented or dementia.

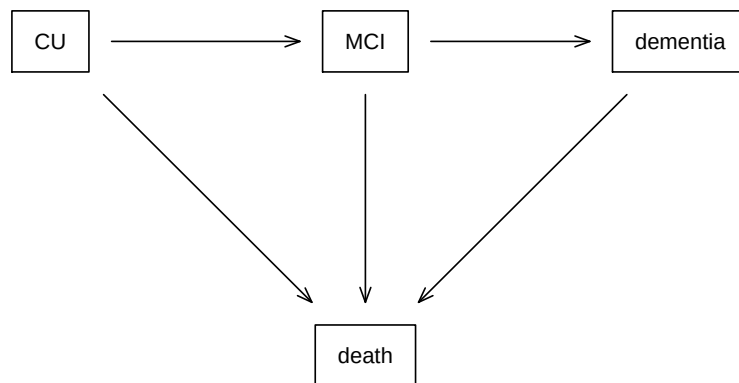


Figure 7: State space model for the dementia data.

The modification of the likelihood is simple. At such a visit, if the diagnosis is “dementia”, then all elements of $p(t)$ for that subject, except state 3 (dementia) are set to 0. For a non-demented diagnosis, elements 3 and 4 (dementia and death) are set to 0, the other two are unchanged. We treat the off study diagnosis as a *marker*, our label for an imperfect assessment of the underlying state. The data will have three distinct outcome variables: death, observed state at a study visit, and partial state at an ordinary office visit.

```
survcheck(Surv2(age, dx) ~1, id=ptnum, data = dementia)
```

Call:

```
survcheck(formula = Surv2(age, dx) ~ 1, data = dementia, id = ptnum)
```

Unique identifiers	Observations	Transitions
4317	21354	4528

Transitions table:

	to	CU	MCI	dementia	nondem	off-dem	death
from	CU	0	845	30	815	184	11
	MCI	449	0	202	257	246	19
	dementia	0	29	0	0	0	172
	nondem	0	0	0	0	0	941
	off-dem	0	0	0	0	0	328
	death	0	0	0	0	0	0

from	to	(censored)
CU		2347
MCI		228
dementia		30
nondem		81
off-dem		102
death		0

Number of subjects with 0, 1, ... transitions to each state:

state	count	0	1	2	3	4	5	6	7	9
CU		3912	363	40	2	0	0	0	0	0
MCI		3560	649	101	5	2	0	0	0	0
dementia		4094	214	9	0	0	0	0	0	0
nondem		3245	1072	0	0	0	0	0	0	0
off-dem		3887	430	0	0	0	0	0	0	0
death		2846	1471	0	0	0	0	0	0	0
(any)		2296	334	1125	380	123	47	9	2	1

```
survcheck(Surv2(age, dx, repeated=TRUE) ~1, id=ptnum, dementia)$transitions
```

from	to	CU	MCI	dementia	nondem	off-dem	death
CU		13030	845	30	815	184	11

MCI	449	951	202	257	246	19
dementia	0	29	163	0	0	172
nondem	0	0	0	2	0	941
off-dem	0	0	0	0	0	328
death	0	0	0	0	0	0

to

from	(censored)
CU	2234
MCI	222
dementia	30
nondem	79
off-dem	102
death	0

As described in the help file for the data set, the on study, off study and death data have been collapsed into a single variable `dx`. The above table of transitions shows that it not uncommon for a patient's diagnosis of MCI at one visit to revert back to CU at a later one. If we fit a model corresponding to Figure 7 using this raw data, the result will be an estimated likelihood of 0, whic would be correct since the data contains an impossible (per the diagram) transition.

A message which will be repeated multiple times is that you need to create the transitions table and *read* it. Really read it, not just a glance. There are several things we can note from the above.

- The `survcheck` routine was written for counting process data. One aspect of `Surv2`, which translates from timeline form to that form, is the repeated argument. When a subject has two sequential visits in the same state, the `survfit` and `coxph` routines need to know whether to treat the second observation as a new event. Most of the time they should not, repeated infections in the CGD data set is a counterexample, however. In this data each row is a visit, the repeated form clearly shows that over half of all visit pairs are CU:CU; most patients stay stable for many years before progression. The (any) row of the total visits table shows that 53% (2296/4327) have no recorded transtions during follow-up.
- There are 30 instances of a CU:dementia pair. Per the model we assume that these subjects passed through MCI, we simply did assess them often enough to observe the transient MCI state.
- Dementia:MCI is much less common than MCI:CU. This reflects on one hand the clinical reality that CU and MCI are more difficult to distinguish than dementia, and also fewer overall dementia cases.
- There are no transitions from death to another state, or from death to censored (further follow-up after death). This is as it should be.
- Off study dementia only transitions to death. This is because further record reviews are not scheduled for demented subjects.
- There are no transitions from nondem to nondem or off-dem. This is, it turns out, an interesting artifact of how the online data collection system was designed. The nurse

abstracters who review the history have a screen with fields for “last assessment date for dementia’ and ’dementia yes/no’. At a second, third, etc review for a patient, should one occur, the contents of these are overwritten; thus the master data set at any given time only has the most recent value.

- More dementias were found in record review than in study visits. We believe that this reflects informative censoring: subjects whose cognition is slipping selectively drop out, i.e., exactly those most likely to receive a dementia diagnosis at their next visit.

Every real study has unique features and patterns like this. Something that looks odd, e.g., item 6 above, needs to be thought through: the “feature” you are seeing may often be your own programming error. The author’s experience has normally been that the total number of calls to survcheck (and other tabulations) while building a proper analysis data set will exceed the number of model fits performed when using that data set.

For this desired analysis we need to create two new variables

- Sequential progression: the first diagnosis of MCI, dementia (or death)
- Off study diagnosis

The first is treated as a known state, the second as a marker. Misclassification will be explored later in the document.

```
table2 <- function(...) table(..., useNA= "ifany") # show NA in tables

noback <- function(x, id, censor) {
  # turn reverse transitions into 'stayed the same'
  # assume the data is sorted by time within unique id, each unique id
  # is a block (but id labels not necessarily in sorted order)
  ix <- ifelse(is.na(x), -1, as.numeric(x)) # placeholder
  if (!missing(censor)) ix[x==censor] <- 0 #make it the smallest level
  new <- unlist(tapply(ix, match(id, unique(id)), cummax))
  # A censored row can occur any time, not subject to the "only up" rule
  if (!missing(censor)) {
    if (is.factor(x)) new[x==censor | is.na(x)] <- match(censor, levels(x))
    else new[x==censor | is.na(x)] <- censor
  } else new[is.na(x)] <- NA

  if (is.factor(x))
    new <- factor(new, 1:length(levels(x)), levels(x))
  else new
}

# first restrict the set of states, then stop backwards transitions
temp <- factor(dementia$dx, c("alive", "CU", "MCI", "dementia", "death"))
ddata <- fromtimeline(Surv2(age, temp, repeated="first") ~
  ptnum + male + educ + any.e4 + cmc,
```

```

data=dementia, id=ptnum)

ddata$maxdx <- noback(temp, ddata$ptnum, "alive")

Error in 'tapply()':
! arguments must have same length

ddata$offdx <- factor(as.numeric(ddata$dx), 5:6, c("ND", "dementia"))

Error in '$<-.data.frame':
! replacement has 0 rows, data has 21354

with(ddata, table2(dx, maxdx))

Error:
! object 'dx' not found

with(ddata, table2(dx, offdx))

Error:
! object 'dx' not found

ddata$iage <- floor(ddata$age)

Error in 'floor()':
! non-numeric argument to mathematical function

# The missclassification table for off study dx
missmat <- matrix(c(1,1,0,0,0,1), ncol=2,
                  dimnames=list(c("CU", "MCI", "dementia"), c("ND", "dementia")))
missmat
      ND dementia
CU      1      0
MCI      1      0
dementia 0      1
# The true transition matrix
mstate <- c("CU", "MCI", "dementia", "death")
mmat <- matrix(0, 4, 4, dimnames=list(mstate, mstate))
mmat[1,2] <- mmat[2,3] <- 1
mmat[1:3,4] <- 1
mmat
      CU MCI dementia death
CU      0  1      0      1
MCI      0  0      1      1
dementia 0  0      0      1
death    0  0      0      0
# Initial absolute rate that is "in the ballpark"
drate <- survexp.us[75, 1, "2010"]* 365.25 # death rate per year, 75 year old
icoef <- rep(log(drate),5)

```

```

dfit0 <- cmsh(Surv(age, maxdx) ~ 1,
              data= ddata, id= ptnum, qmatrix= mmat,
              init = icoef, exact="death",
              marker= list(state("CU", "MCI", "dementia"):offdx ~0
                           /discrete(init=missmat)))

```

```

Error:
! object 'age' not found

```

```

dfit0$fit$iter

```

```

Error:
! object 'dfit0' not found

```

```

dfit0

```

```

Error:
! object 'dfit0' not found

```

```

# Add covariates
dfit1 <- cmsh(list(Surv(age, maxdx) ~ iage,
                  1:2 + 2:3 ~ any.e4,
                  0:4 ~ male /common),
              data= ddata, id= ptnum, qmatrix= mmat,
              init= dfit0, exact= "death",
              marker= state("CU", "MCI", "dementia"):offdx ~0
                           /discrete(init=missmat)
              )

```

```

Error:
! object 'age' not found

```

```

dfit1$fit$iter

```

```

Error:
! object 'dfit1' not found

```

```

dfit1

```

```

Error:
! object 'dfit1' not found

```

There is a lot to unpack in the above example.

- The use of first observed MCI and first observed dementia as our endpoint simplifies the model. This endpoint may be of medical interest in and of itself, as it corresponds perhaps more closely to clinical practice than the “true MCI” we will estimate a little later. The first diagnosis of MCI may for instance be the basis for treatment decisions.
- The (declared to be) true transition matrix, `mmat` above, guides the underlying model fit.

It is not optional.

- The formula statement is a list, the first element of the list is the default formula, which serves as the starting point for all model transitions (the 5 arrows in Figure 7). Subsequent rows modify the formulas for one or more of those five. 1:2 stands for the transition from state 1 to state 2, numbers correspond to the rows/cols of the transition matrix `mmat`, 'CU': 'MCI' is equivalent to 1:2. 0:5 is shorthand for “any state”:5, in this case all 3 transtions to death have male sex as a covariate, and all share the same coefficient.
- The default initial value of 0 for all coefficients may often not work for the intercepts. It corresponds to a mean time to transition of $\log(2)/1 = .7$ units of time. If the time axis were in days, for instance, the log-likelihood at $\beta = 0$ can be so small that the `exp()` function underflows, giving a probability of exactly 0 and correponding log-likelihood of $\log(0)$. The routine may occasionally recover from such a bad starting point, or may not.

The `marker` statment gives a mapping between a subset of states and a measured but imperfect marker of those states. The form `state:marker` reflects that underlying probability model is based on $\Pr(\text{observed value} \mid \text{state})$, not $\Pr(\text{state} \mid \text{observation})$. The missclassification matrix `mmat` has three rows and each row is a probability distribution, each has to add to 1. We read it as

- If the true state is CU and the marker `offdx` is observed, then $\text{probability}(\text{offdx} = \text{“ND”}) = 1$
- If the true state is MCI and the marker `offdx` is observed, then $\text{probability}(\text{offdx} = \text{“ND”}) = 1$
- If the true state is demenentia and the marker `offdx` is observed, then $\text{probability}(\text{offdx} = \text{“ND”}) = 0$
- There is no row for death, for two reasons. The first is that if death is observed at a given date, we will not have observed a nondemented/demented status on that same date, thus the case does not arise. The second is to ask what numerical value one would put in that row, given the constraint that it should add to 1?
- It is of course possible for someone to have had an office visit and evaluation, then die in a car accident that same day on the way home. When there is a tied time like this, our experience is that $P(\text{data error})$ is by far the most likely case. If such a case did occur, then to avoid complaints from the code the user will need to break the tie, e.g., make the office visit .1 day earlier or some such.

The `, 0` portion of the formula states that there are no estimated coefficients for this markere, and is followed by the distribution and parameters of the distribution.

The `msm` package takes a different approach to the case of partially observed states as shown below.

```
istate <- with(ddata, ifelse(is.na(offdx), as.integer(maxdx) - 1L,
                             c(5,3)[as.integer(offdx)]))
```

```

Error:
! object 'offdx' not found

table2(ddata$dx, istate) #1,2,3,4 have to line up with qmat

Error:
! object 'istate' not found

qmat <- mmat
qmat[qmat>0] <- drate # initial values for the intercept
otype <- ifelse(ddata$dx=="death", 3,1)
mfit0 <- msm(istate ~ age, data=ddata, subject= ptnum, death=4,
             qmatrix=qmat, obstype=otype,
             censor=c(0,5), censor.states=list('0'= 1:3, '5'=1:2))

Error:
! object 'age' not found

rbind(cmsh= exp(coef(dfit0)), msm= mfit0$Qmatrices$baseline[qmat>0])

Error:
! object 'dfit0' not found

# essentially the same solution

```

The censor and censor.states arguments identify outcomes in istate that map to multiple states, and everything else is assumed to be exact. This approach does not extend to more general missclassification matrices.

Over half the compute time for cmsh is spent in the matrix exponential. For a small set of state patterns msm has coded special routines for the matrix exponential, and in those cases is can be significantly faster than cmsh.

5.1 CAV example

Data from a study that monitors the progression of coronary allograft vasculopathy (CAV) is provided in the msm package. CAV is a post-transplant deterioration of the aterial walls in patients who have recieved a heart transplant. Make a few changes to the data set to set it up for the cmsh function, and show some summaries. First fit CAV outcome as interval censored.

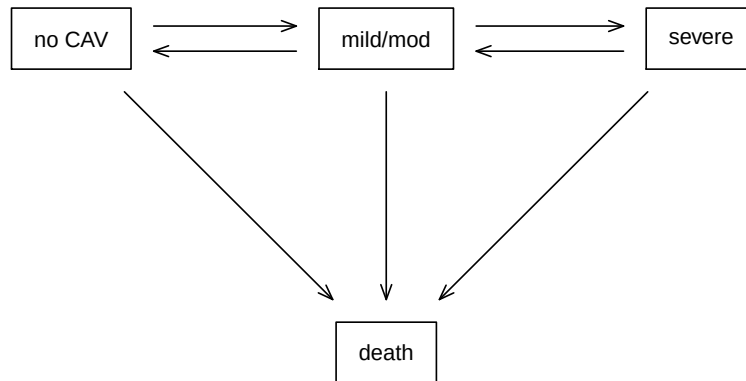
```

library(msm)
cav2 <- cav
names(cav2) <- casefold(names(cav2)) # lower case (personal preference)

# The model
cstates <- c("no CAV", "mild/mod", "severe", "death")
cqmat <- matrix(0, 4, 4, dimnames=list(cstates, cstates))
cqmat[1,2] <- cqmat[2,1] <- cqmat[2,3] <- cqmat[3,2] <- 1
cqmat[1:3,4] <- 1

```

```
statefig(rbind(3,1), cqmat, offset=.02)
```



```
# multi-state status for Surv(), first level has to be "no event at this time"
cav2$cstate <- factor(cav2$state, 0:4,
                      c("alive", "no CAV", "mild/mod", "severe", "death"))
table(cav2$cstate)      # no off study FU!
```

```

  alive  no CAV mild/mod  severe  death
    0    2039    351    205    251
survcheck(Surv2(years, cstate) ~1, cav2, id= ptnum)
```

Call:

```
survcheck(formula = Surv2(years, cstate) ~ 1, data = cav2, id = ptnum)
```

Unique identifiers	Observations	Transitions
622	2224	616

Transitions table:

from	to				
	no CAV	mild/mod	severe	death	(censored)
no CAV	0	204	44	148	269
mild/mod	46	0	54	48	30
severe	4	13	0	55	16
death	0	0	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count					
state	0	1	2	3	4	5
no CAV	576	42	4	0	0	0
mild/mod	430	168	23	1	0	0
severe	530	86	6	0	0	0
death	371	251	0	0	0	0
(any)	258	210	78	59	12	5

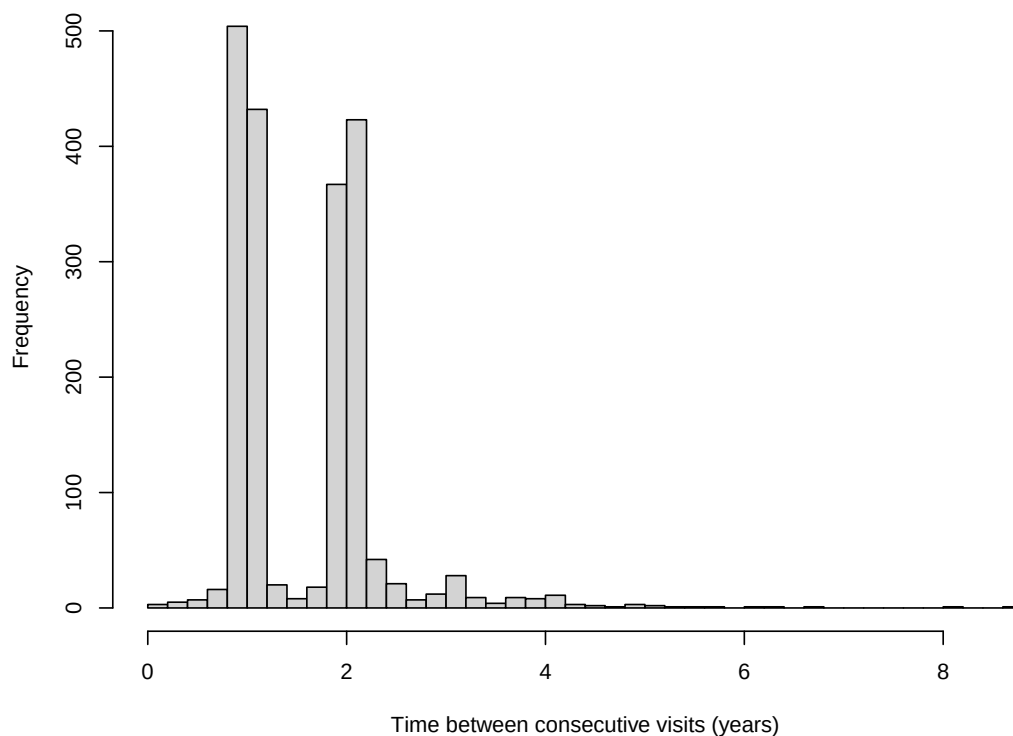
table(table(cav2\$ptnum)) # number with 1, 2, 3, ... visits

2	3	4	5	6	7	8	9	10	11	12	13	14	15
169	129	70	60	40	57	40	23	21	3	3	3	3	1

We use a standard multi-state `Surv(time, state)` form of the survival package, where the state variable is a factor such that the first level of the factor represents “no event at this time”. The CAV data is unusual in that there are no censored observations, each interval ends in one of the 4 states. The 622 subjects have between 2 and 15 assessments, 27% with 2 and 21% with 3. The transitions table above shows the count of observed transition pairs, a subject with 5 visits will contribute 4 counts to the table. Remaining in the no CAV state accounts for the largest number of visit pairs. As should be the case, there are no transitions from death to another state.

As seen below, the inter-visit intervals vary widely, a situation that favors an interval censored model.

```
# what are the inter-visit intervals?
i1 <- duplicated(cav2$ptnum, fromLast=TRUE)
i2 <- duplicated(cav2$ptnum)
interval <- cav2$years[i2]- cav2$years[i1]
ideath <- cav2$state[i2]=="death" # did the interval end in death?
#between visits
hist(interval[!ideath], nclass=50, main=NULL,
      xlab="Time between consecutive visits (years)")
```



```
#hist(interval[ideath], nclass=30, main=NULL,
#       xlab="Years between last visit and death")
```

For interval censored data we need to specify the set of valid transitions Q . This is a matrix of 0/1 values, which can also be used by the `statefig` routine. The row/column names of the transition matrix are the names of the states, levels in the `Surv` object need to match these.

```
cqmat
      no CAV mild/mod severe death
no CAV      0      1      0      1
mild/mod    1      0      1      1
severe      0      1      0      1
death       0      0      0      0
cfit0 <- cmsh(Surv(years, cstate) ~ 1, data= cav2, id= ptnum,
              qmatrix= cqmat, exact="death")
cfit0 #msm vignette, 2.5, model 1
Call:
```

```
cmsh(formula = Surv(years, cstate) ~ 1, data = cav2, id = ptnum,
      qmatrix = cqmat, exact = "death")
```

```

      2:1  1:2  3:2  2:3  1:4  2:4  3:4
(Intercept) -1.49 -2.06 -2.04 -1.07 -3.16 -3.21 -1.18
```

```
States: 1= no CAV, 2= mild/mod, 3= severe, 4= death
```

```
Log-likelihood: ( -5395.578, -1984.399 ), 2*diff: 3411.2 on 7 df
2846 observations 622 subjects
```

For the above model states are of two types: the state is known at the measurement time but not when that state was entered, and exact states, where the entry time to the state is known. These latter are enumerated in the **exact** argument. Death is the most common exact state. One can also have multiple exact states, one of the more common examples is cause of death, e.g., states of 'death due to cancer' and 'other death' had been separated by the user. The status variable in the `Surv` call is a factor with censored as the first level, followed by the states, in the same order as the Q matrix. Now add covariates.

```
cfit1 <- cmsh(list(Surv(years, cstate) ~ 1,
                  1:2 + 1:4 ~ sex),
              data=cav2, id= ptnum, init= cfit0,
              qmatrix= cqmat)
cfit1 # msm vignette, 2.7, model 2a
Call:
cmsh(formula = list(Surv(years, cstate) ~ 1, 1:2 + 1:4 ~ sex),
      data = cav2, id = ptnum, qmatrix = cqmat, init = cfit0)
```

```

      2:1  1:2  3:2  2:3  1:4  2:4  3:4
(Intercept) -1.486 -1.991 -2.037 -1.073 -3.177 -3.214 -1.18
sex          0.000 -0.657  0.000  0.000  0.147  0.000  0.00
```

```
States: 1= no CAV, 2= mild/mod, 3= severe, 4= death
```

```
Log-likelihood: ( -1984.399, -1980.068 ), 2*diff: 4.3 on 9 df
2846 observations 622 subjects
```

The code allows the use of a prior model for the initial values. Interestingly, the number of iterations needed to estimate the initial intercept was larger than for **cfit1**. This is not uncommon, in our experience.

```
cfit2 <- cmsh(list(Surv(years, cstate) ~ dage,
                  1:2 + 2:3 ~ sex / common,
                  2:1 + 3:2 ~ sex / common,
                  0:death ~ sex / common), data=cav2, id= ptnum,
```

```

      qmatrix= cqmat, exact="death", init=cfit0)
cfit2
Call:
cmsh(formula = list(Surv(years, cstate) ~ dage, 1:2 + 2:3 ~ sex/common,
  2:1 + 3:2 ~ sex/common, 0:death ~ sex/common), data = cav2,
  id = ptnum, qmatrix = cqmat, init = cfit0, exact = "death")

              2:1      1:2      3:2      2:3      1:4
(Intercept) -1.34154 -2.51624 -1.84722 -0.60829 -4.32545
dage         -0.00473  0.01832 -0.00674 -0.01461  0.03561
sex           0.29771 -0.42289  0.29771 -0.42289  0.37266

              2:4      3:4
(Intercept) -1.38358 -0.81
dage         -0.06235 -0.01
sex           0.37266  0.37

States:  1= no CAV, 2= mild/mod, 3= severe, 4= death

Log-likelihood: ( -1984.399, -1960.838 ), 2*diff: 23.6 on 17 df
2846 observations 622 subjects
cfit2$cmap
              2:1  1:2  3:2  2:3  1:4  2:4  3:4
(Intercept)    1    4    7    9   11   14   16
dage            2    5    8   10   12   15   17
sex             3    6    3    6   13   13   13

```

More complex models are specified using a list for the formula argument. The first formula in the list specifies the response and a default set of covariates which is used as the starting point for all transtitions. In the above example age at diagnosis `dage`. Later formulas update the formula(s) for specified transtions

- line 2 of `cfit1` adds sex to the formulas for none to mild (1:2) and none to death (1:4), but not to other transitions
- line 2 of `cfit2` forces the the 1:2 progression (no CAV to mild/moderate) and 2:3 (mild/moderate to severe) to share a common sex coefficient
- line 3 of `cfit2` makes the sex coefficients for the two reverse transitions identical
- 0:death stands for any state to death. Transitions marked as 0 in `qmatrix` are ignored whenever exanding a shorthand like this, i.e., a death:death transtion

The `cmap` component of the returned fit is the overall map between covariates, linear predictors, and coefficients. The final fit has 17 unique coefficients, the linear predictor for the 3:4 (severe:death) transition for example makes use of coefficients 16, 17 and 13.

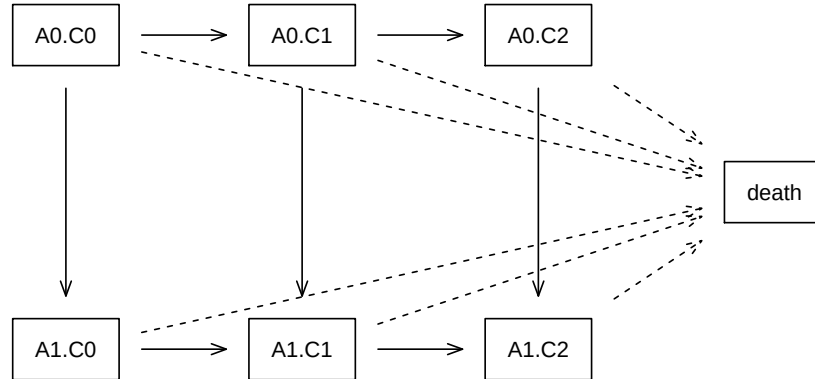
6 What is a hidden Markov model (HMM)?

The answer to this is that HMM is a general technique, and we see the label applied to many different problems in many different subfields. The `cmsh` package addresses a particular structure, one that has been useful in our analyses of data from the Mayo Clinic Study of Aging (MCSA).

Our models can be thought of having two levels. Level 1 is a multi-state hazards model, much like those addressed by multistate hazard (MSH) models fit using the `coxph` function in the survival library. Here is a moderately complex model for illustration, based on a dementia study.

```
states <- c(outer(paste0("A", 0:1), paste0("C", 0:2), paste, sep='.'), "death")
statedata <- data.frame(state= states,
                        A=c(0,1, 0,1, 0,1, 2),
                        N=c(0,0, 1,1, 2,2, NA),
                        D=c(0,0, 0,0 ,0,0, 1))

statedata
  state A  N D
1 A0.CO 0  0 0
2 A1.CO 1  0 0
3 A0.C1 0  1 0
4 A1.C1 1  1 0
5 A0.C2 0  2 0
6 A1.C2 1  2 0
7 death 2 NA 1
qmat <- qplus(statedata, death= "death")
statefig(cbind(2,2,2,1), qmat, alty=rep(1:2, c(7,8)))
```



The above model assumes two latent processes, amyloid accumulation in the brain, A0 and A1, and cognitive decline from no impairment to high, C0, C1, C2. The premise of an HMM is that these states are not directly observed, rather what we see are biomarkers that indicate, imperfectly, the underlying process. The latent or “hidden” states operate in continuous time, so transitions in the figure are a change in A or N, never both at once. Because of intermittent observation a subject might well be in A0:N0 at one visit and A1:N2 at the next, but the unobserved continuous process would not have made an instantaneous multistate jump.

Here is a possible specification for a model. It is purposely a very general one.

```
mfit <- cmsh(list(Surv(time, death) ~ male,
  0:death ~ iage,
  A(0):A(1) ~ apoe + male + ns(iage,4)/ common,
  C(0):C1(1) + C(1):C(2) ~ iage + cmc),
marker = list(C:dx ~ 1/ multinomial(pattern= pmat),
  C:memory~ 1 /logistic,
  C:memory~ practice/ logistic(param=mean) + common,
  A(1:2):centiloid + A(1:2):sqrt(p.tau181) ~1/gaussian),
```

```
data= dementia, id=ptnum, qmatrix = qmat,
statedata = statedata)
```

Full estimation includes 13 transition rates (arrows), along with the relationship of 4 biomarkers (dx, memory, centloid, and p.tau181) to the underlying states. It is a complex model, with a lot of parameters, and the specification is also complex. The statedata data set makes specification a bit easier by setting up aliases, e.g., A(0) is shorthand for A0:C0, A0:C1 and A0:C2. The data set has a row per state, in the same order as they are found in the transition matrix `qmat`. It's first column must be a variable named "state", the other columns provide aliases.

The specification for the 13 transitions rates has the same form as in the `coxph` function. Each of the rates is modeled as $\exp(\beta X)$, in the same manor as a Cox model, each transition has it's own equation. The formula argument, the first argument of the `cmsh` function, is a list whose first element is the default formula for each transition, subsequent lines each update the formula for selected transitions. Here is a sample from the above:

- All 6 transitons to death are of the form `male + iage`; each has separate coefficients
- The A0C0:A1C0, A0C1:A1C1 and A0C2:A1C2 transitions, the three vertical arrows in the figure, have the form `male + apoe + ns(iage,4)`; these three transitions have the same coefficients.
- The four C transitions (horizontal arrow) have form `iage + cmc`

Behind the scenes the A(0):A(1) specification translates as any valid transition from a state where the A column of `statedata` is 0, to one where the A column is 1. Going from row 1 (A0.C0) to row 6 (A1.C2) formally satisfies the A(0) to A(1) pattern, but is not one of the valid transitions, per `qmat`.

The marker specification is also a list of formulas. The `C:memory /gaussian` specification states that each of the true cognitive states C has an expected distribtuion of memory scores which follows a logistic distribution: true state C(0) gives rise to $\text{logistic}(\mu_1, \sigma_1)$ memory scores, true state C(1) gives rise to $\text{logistic}(\mu_2, \sigma_2)$ memory scores, and true state C(3) to $\text{logistic}(\mu_3, \sigma_3)$. This gives rise to 6 linear predictors in the model, one for each mean and $\log(\text{std})$ value. Memory scores tend to be systematically better when the same test is readministered a second time, this practice effect is often coded as a factor with levels of 0, 1-2, 3+ for the number of prior tests. The second memory line states that the linear predictors for $\mu_1 - \mu_3$ will also include the practice effect, and that this effect is assumed to be the same size for all 3 latent states. (An assumption that is open to challenge; a C2 subject might not remember prior test exposures as well.)

A latent state of none/low amyloid, A(0), is assumed to lead to gaussian distributions for centloid values and for $\log(\text{tau})$, PET based biomarkers, and ditto for A(1). The A(3) row, death, will never correspond to a PET measurement. This leads to 2 markers * 2 state groups * 2 parameters = 8 linear predictors. In this case each LP is simply an interecept. The diagnosis variable is categorical, leading to a multinomial distribution for the state:marker combination. The pattern argument will be illustrated in a later example.

The first major difference between the MSH and HMM models is that observations on each state come from interval censored data, also known as panel data. That is, subjects visit at regular intervals and undergo a panel of tests. We will know their state at a given test date,

but not the time at which the study participant transitioned to that state. This leads to a more complex likelihood than a MSH model, but also more realistic for data such as this.

The second difference is that an HMM treats the states as latent factors. The model will have standard covariates, things like age, sex and genotype which affect the transition rates from state to state, and also reporter or biomarker variables on the states themselves. In this case, we might

	True	Observed			
		CU	MCI	dementia	death
have the following misclassification matrix.	C0	$1 - p_1$	p_1	0	0
	C1	p_2	$1 - (p_2 + p_3)$	p_3	0
	C2	0	p_4	$1 - p_4$	0
	death	0	0	0	1

We have been careful above to use one set of labels for the true states (C0, etc) and another for the observed states. In many cases a user will be tempted to use the same labels for the true underlying state and for the clinical observation: this is a good way to generate confusion, and we recommend that you do otherwise. Death is a special state, it is not interval censored as we know the exact death date, nor is it a latent state since it is observed without error.

A primary reference for the material is Christopher Jackson's excellent manual for the msm package [5]. Others that I found particularly helpful were a web page by Nikolai Shokiriv [10], Satten and Lingini [9] and computational details from Kalbfleisch and Lawless [6].

Per Shokhirev there are 3 canonical problems for HMM models.

1. Given the model parameters, compute the probability of a particular output sequence. This is solved by the Forward or Backwards algorithms.
2. Given the model parameters and an output sequence, compute the most likely path for the hidden states (the true but unobserved path). This is solved by the Viterbi algorithm.
3. Given an output sequence, find the most likely set of model parameters. This is solved by the Baum-Welch algorithm.

We are focused on problem 1: an underlying search algorithm will choose parameters β , which map to rates, which in turn lead to a likelihood value. The search algorithm uses this to find an MLE. Many references, e.g. Wikipedia, focus only on problem 2 the exclusion of the others. This led to much early confusion for this author, e.g., the Viterbi algorithm involves a forward recursion and earns a 'forward' label as well. It might look like 3 is our target, but that would only be true if there were no covariates and we were directly estimating the rates.

Further math and computational details are found in a mathematical vignette.

7 Design

Specification of an HMM model is complex, because the model is complex. We need to specify

- The underlying Markov model: the states, state names, and the allowable transitions.
- Which covariates apply to each transition, and which of these transition/covariate pairs share a coefficient.

- For each marker y transmission function $P(y|s)$ needs to be defined, which provides the distribution of y given the true state s . (Death does not need a transmission function).
- The initial probabilities, whether these should be estimated, and if so what covariates to use.
- Initial values for the coefficients and specification of which of them, if any, are fixed.

One of the attractions of the HMM likelihood is that reporter variables need not all be present. In the MCSA for instance, visits are scheduled every 15 months, but PET imaging is only gathered on a subset, and for many of those PET is done every 30 months. For the HMM likelihood there is no need to “fill in” those other measurement using imputation or other special missing value computations, etc. Each measured reporter adds to the likelihood and, those which are not measured at this visit don’t. Simple.

8 Time dependent covariates

Though markers for the states can be observed intermittently, covariates that affect the rates can not be missing, the same restriction that applies in a MSH model. If cerebral vascular disease (CeVD) were the state, for instance, with an intermittently observed predictor of the disease rate such as systolic blood pressure, the SBP would commonly be filled in wrt continuous time by using the most recently measured value, also called last value carried forward; see the `lcvf` function.

There are 3 kinds of states in the `hmm` fitter: latent states, which are not observed precisely but for which we have observed but imperfect *markers* of the current state, interval censored states, where the state a subject is in at the observation time is certain, but not when they entered the state, and exact states where both the state and its time of entry is known. Both interval censored and exact states should appear as levels in the in the second argument to `Surv`, `cstate` above, which must be a factor whose first level corresponds to “no observed state at this time” and remaining levels will match the state labels in the Q matrix. (Common labels for this first level are “censor” or “none”). Death is the most common exact state, so much so that “death” is the default value of the `exact` argument, and so we could have omitted it from the above call.

The above fits have no latent states, so the observed outcome is captured in the `Surv` argument. The `formula` argument specifies the models for the between state transitions, and has the same form as in the `coxph` function of the survival package. It is a list where the first formula is the default, the left hand side is the response and the right the covariates. Further elements of the list have the form of one or more state:state transitions on the left and covariates on the right. Behind the scene there is a formula for each transition, each starts as the default and is modified by each later addition or subtraction, using the same logic as `update.formula`. The `common` option states that the specified transitions share a common `sex` coefficient. The result of the formulas is captured in `cmap` component of the fit, each positive integer is an index into the overall coefficient vector. The following are valid specifications of transitions:

- `0:death, 0:state('death'), 0:4`: transition from any other state to death. Illegal transition are silently removed, i.e., this does not include death:death.

- `c(1:3):4, state('no CAV', 'mild/mod', 'severe'):death`: same as the above
- `''no CAV'':''mild/mod'' + ''mild:mod'':''severe''`: same as 1:2 + 2:3

The default for `cmsh` is to not maximize, returning the result at the initial parameters. The second two calls use what we have found to be the most robust option for general practice, which is based on a Levenburg-Marquart algorithm along with a trust region, and uses the outer product of the score vectors U_i per subject as an approximate Hessian matrix.

9 Example data

We will use the included `dementia` data set as a running example for the remainder of the vignette. This data set is a simulant, based on data from the Mayo Clinic Study of Aging which was used in the analyses of [4] and [3]. The MCSA is an age and sex stratified sample of subject from Olmsted County, Minnesota. After recruitment, subjects are invited to return for a study visit every 15 months, with the goal of understanding the evolution of cognitive decline along with various markers of the pathology that underlies such decline. The `dementia` data set focuses on two processes. The first is cognitive loss, as measured by clinical assessment, cognitively unimpaired (CU) versus mild cognitive impairment (MCI) versus dementia, a memory score based on a battery of cognitive tests, and hippocampal volume, a measure of neural loss that presumably underlies cognitive decline. The second is progression of amyloid and tau deposition, known to be part of the Alzheimers disease pathway, direct PET imaging of amyloid and tau in the brain, and a putative blood based biomarker P-tau181 of amyloid load. (Whether amyloid and tau are causative the disease or only convenient markers process is vigorously debated.) Death is a competing risk for both processes.

10 Multi-state hazard model

We first fit a multi-state hazard (MSH) model, using functions from the survival package. This will help make the data clearer, along with the differences between MSH, interval censored and HMM models. The MSH fit will partially reprise the analysis in [4], which fits the simple 3 state model of not demented (ND)/dementia/death on age scale, with sex, amyloid, CMC, and APOE positivity as covariates.

The choice of time scale is a critical step in the analysis plan. One useful way to think about this is as a choice of which covariates to explicitly model and which covariates to adjust for by matching. The advantage of matching on some covariate x is that it provides the best possible adjustment: it works independently of the true functional form of the relationship of the relationship $y = f(x)$. A disadvantage is that we do not get a p-value, or an estimate of f . In epidemiology studies of chronic disease, matching on age and sex is often a natural choice: we already know that rates rise with age and differ by sex, no one needs a p-value “proof” of this, and a given study is usually focused on other questions. In a simple Cox model or a multistate hazard (MSH) model, the partial likelihood is based on matching. If time-since-enrollment is used as the underlying scale then for an observed dementia in subject “Smith”, say, at 249 days post enrollment, Smith is compared to all others who were also alive and at risk at 249 days post enrollment, using their appropriate covariate values at that time. If age is used as the time scale

and that same event happened when Smith was 30104 days of age (approx 82y 8m), the matched group will be all those who were active in the study at 30104 days of age, some of whom might have been enrolled in the study for a month while others had been active for a decade. A general rule of thumb for the Cox model is to match on the time variable that is the most impactful and/or difficult to model, and model the easier and/or less impactful time scale. I often tell the researcher “Assume you want to predict a subject’s risk over the next year, and you can have only one of age, time since first diagnosis of disease, or time since enrollment in the study to guide you. Which would you choose?”

Both death and dementia rates rise rapidly with age, death rates are approximately log-linear from age 50 onward but dementia rates have a different form (a later but more precipitous rise). The MCSA is an age and sex stratified random sample from a defined population, enrollment ages range from 50 to 90 and follow-up of up to 16 years. The data thus has observations from 50 to 100+ years of age, over this range population death rates increase by approximately 150 fold, e.g., `survexp.mn[100,, '2020'] / survexp.mn[50,, '2020']`. If modeling the effect of age on the hazard one really needs to get it right, blindly adding `+ age` to the formula will not suffice, since even a small error can cause effects of interest, such as the 1.3–1.4 fold effect APOE genotype, to be completely lost in the resulting bias. On the other hand, one would expect little clinical effect for time since enrollment, which is the time since a subject was randomly selected from the master age/sex registry of Olmsted County residents for an invitation letter. This makes age the natural time scale for an MSH model.

A model using time from enrollment as the time scale would include age as a covariate; similarly a model using age as the time scale should include time-since enrollment as a covariate. Based on prior experience we have coded this as a categorical time-dependent variable of 0-1 years, 1-4, and 4+ years post enrollment, with 4+ as the reference level. The dementia data set included in the package is modeled on the actual MCSA data, results will not match any analyses using the actual MCSA sample but will often be similar in magnitude.

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